



Introduction to Structural Equation Modeling

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UNIVERSITÀ
DI TRENTO
Dipartimento di
Psicologia e Scienze Cognitive

PhD course in *Psychology and Cognitive Science*
Department of Psychology and Cognitive Science
University of Trento
June 4 / 5 / 10 / 12 / 19, 2026



Who am I?

Academic

Researcher in Tenure Track (RTT)

Work and Organizational Psychology (M-PSI/06)

July 1 2024 – ongoing

Research Fellow (RTD-a)

Work and Organizational Psychology (M-PSI/06)

July 1, 2021 – June 30, 2024

Postdoctoral Researcher – University of Trento

Work and Organizational Psychology (M-PSI/06)

Sept 15, 2018 – June 30, 2021

Education

Ph.D. – Sapienza University of Rome

Personality and Organizational Psychology

Feb 2018

M.S. – University of Bologna

Clinical Psychology

Feb 2013

B.S. – University of L'Aquila

Psychology

Oct 2010



Who am I?

Research Areas

1. Organizational Psychology, Personality Psychology, Psychometrics
2. Educational Psychology and Individual Differences

Research Topics

1. Social Desirability
2. Individual differences in personality from childhood (moral disengagement, effortful control) to Adulthood (big five, positivity)
3. Methodological and theoretical issues in the study of self-esteem
4. Personality factors and organizational outcomes (e.g., Well-Being at work)
5. Methodological issues in organizational psychology and management
6. Non-cognitive skills in junior high school students



Who am I?

Research Projects and Practical Applications

1. GdF (Sapienza University of Rome)
2. Mobbing (Autonomous Province of Trento and Univ of Trento)
3. Nudging and Unemployment (Autonomous Province of Trento and Univ of Trento)
4. Non-cognitive skills in Junior high school students (Fondazione Caritro and Univ of Trento)
5. WebeWo LAB -> Interdisciplinary lab (Law, Psychology, Sociology) that aimed to study interventions, antecedents, and outcomes of well-being at work
6. PRIN PNRR 2022 – P.I. of the project «*Loneliness, non-cognitive skills, and academic achievement in Junior High School: Assessment and intervention based on Big Data, longitudinal, and intensive methods*»

More info at

<https://webapps.unitn.it/du/it/Persona/PER0209265/Curriculum>



Who Are You?

....please, introduce your

- *name*
- *background*
- *skills*
- *acquired knowledge*
- *expectations about this course..... and the PhD in general*

This is useful to start creating a collaborative, positive, and multidisciplinary organizational climate...



Schedule

- 04 June 2026 – Aula 11
- 05 June 2026 – Aula 11
- 10 June 2026 – Aula 11
- 12 June 2026 – Aula 11
- 19 June 2026 – Aula 11 (*Project Presentation*)

Your project for this PhD course will involve one of the following options:

- A theoretical or practical argument focusing on a Structural Equation Modeling (SEM) technique (e.g., mediation with SEM, Longitudinal SEM, Latent Growth Modeling, etc.)
- The examination and discussion of one or more articles that report SEM findings.
- Data analysis using *Mplus* or R. You have the flexibility to choose whether to analyze your own dataset, an open-access dataset, or a simulated dataset.
- Hypothesize a SEM pipeline – from Model Formulation to Model Evaluation – accompanied by hypothetical codes (in R or *Mplus*)



Let's start our journey...

Course Material available at
<https://github.com/EnricoPerinelli/Intro-to-SEM>



The need for Latent Variable Models

Problems with *Measurement*

The focus of Psychology is the study of human behaviors and human cognitive processes. Thus, it is difficult to think at tools capable to assess them.



The need for Latent Variable Models

Problems with *Measurement*

In 1940 a committee established by the British Association for the Advancement of Science concluded that measurements taken in psychological methods could not be considered “scientific measurements”.



The need for Latent Variable Models

Solutions for improving *Measurement*

Stevens (1946)

MEASUREMENT

is defined as the
assignment of numerals
to objects or events
according to rules

SCIENCE

Vol. 103, No. 2684

Friday, June 7, 1946

On the Theory of Scales of Measurement

S. S. Stevens

Director, Psycho-Acoustic Laboratory, Harvard University

FOR SEVEN YEARS A COMMITTEE of the British Association for the Advancement of Science debated the problem of measurement. Appointed in 1932 to represent Section A (Mathematical and Physical Sciences) and Section J (Psychology), the committee was instructed to consider and report upon the possibility of "quantitative estimates of sensory events"—meaning simply: Is it possible to measure human sensation? Deliberation led only to disagreement, mainly about what is meant by the term measurement. An interim report in 1938

by the formal (mathematical) properties of the scales. Furthermore—and this is of great concern to several of the sciences—the statistical manipulations that can legitimately be applied to empirical data depend upon the type of scale against which the data are ordered.

A CLASSIFICATION OF SCALES OF MEASUREMENT

Paraphrasing N. R. Campbell (Final Report, p. 340), we may say that measurement, in the broadest sense, is defined as the assignment of numerals to objects or events according to rules. The fact that



The need for Latent Variable Models

Solutions for improving *Measurement*

Stevens (1946)

SCALES
Nominal
Ordinal
Interval
Ratio

TABLE 1

Scale	Basic Empirical Operations	Mathematical Group Structure	Permissible Statistics (invariantive)
NOMINAL	Determination of equality	<i>Permutation group</i> $x' = f(x)$ $f(x)$ means any one-to-one substitution	Number of cases Mode Contingency correlation
ORDINAL	Determination of greater or less	<i>Isotonic group</i> $x' = f(x)$ $f(x)$ means any monotonic increasing function	Median Percentiles
INTERVAL	Determination of equality of intervals or differences	<i>General linear group</i> $x' = ax + b$	Mean Standard deviation Rank-order correlation Product-moment correlation
RATIO	Determination of equality of ratios	<i>Similarity group</i> $x' = ax$	Coefficient of variation

The need for Latent Variable Models

Solutions for improving *Measurement*

Self-report questionnaires, Other-report questionnaires, Objective measures, ecc.

Two pivotal concepts:

RELIABILITY → Precision of a measurement ($X_i = T_i + E_i$)

VALIDITY → The instrument is a proper tool for measuring what it aims to measure.

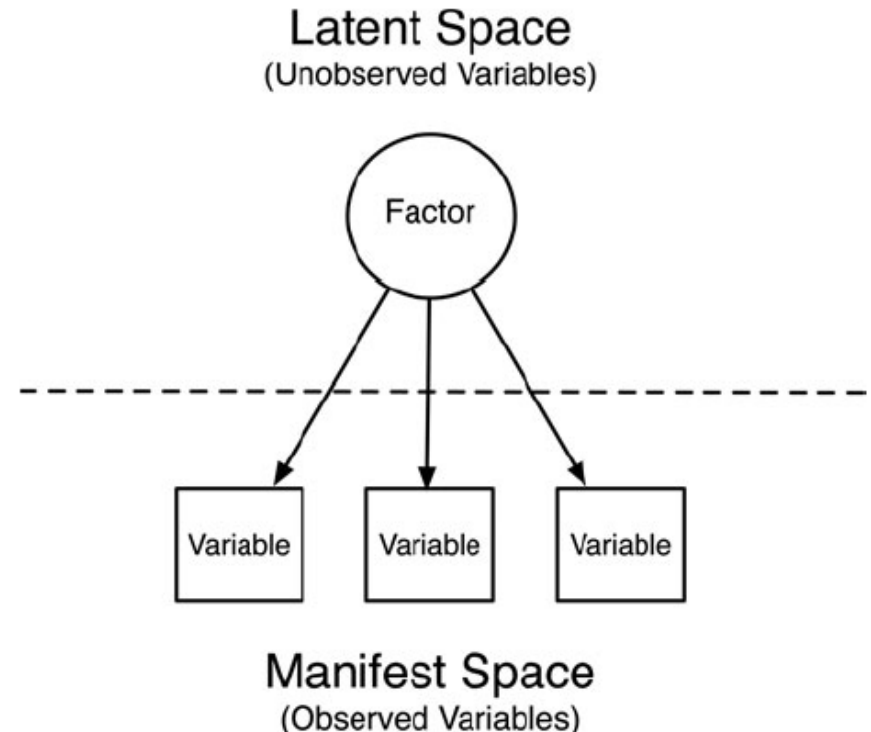
An instrument may be reliable, but not valid (e.g., I cannot use a “reliable” ruler for measuring intelligence)

The need for Latent Variable Models

Solutions for improving *Measurement*

The focus on statistical methods has been pivotal for the advancement of the concept of measurement in Psychology

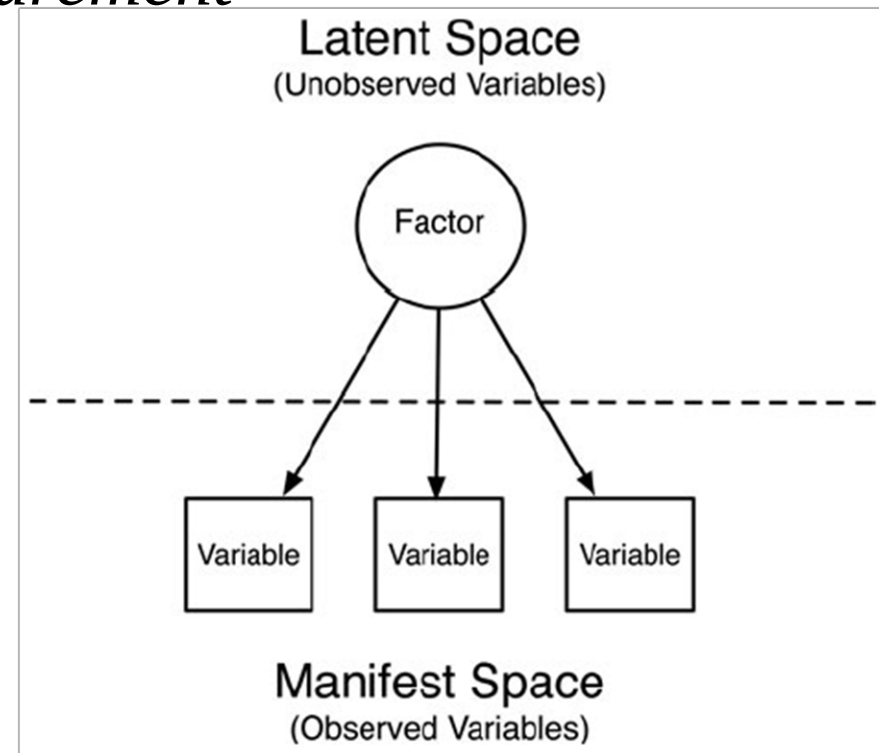
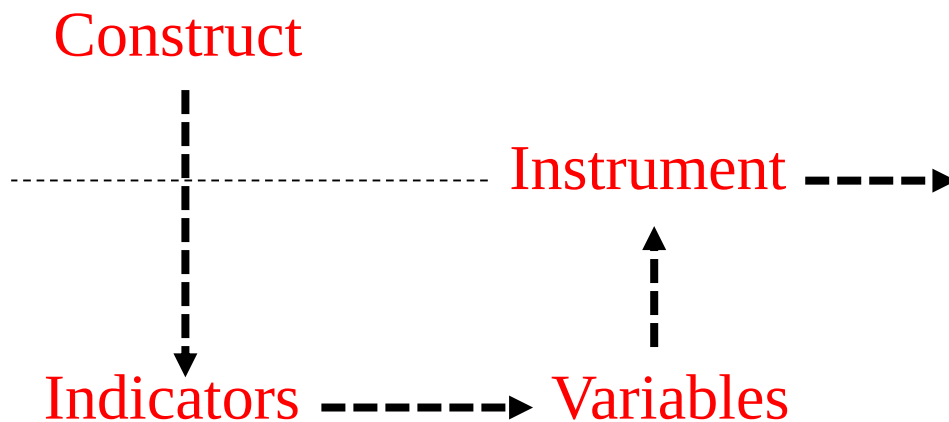
From *Nesselroade & Molenaar (2016)*



The need for Latent Variable Models

Solutions for improving *Measurement*

Operationalization Process



**Operationalization
Process:
An Example**

Conscientiousness

**Big Five
Questionnaire**

(only selected items
that refer to
conscientiousness)

“Scrupulousness”
(orderliness and
precision) and
“Perseverance”
(capability of
fulfilling one’s own
tasks and
commitments)

ITEM_1 *I tend to be very
thoughtful*
ITEM_2 *Before completing
a job I spend a lot of time
revising it*
ITEM_3 *It’s difficult for me
to give up an activity I’ve
undertaken*
ITEM_4 *If I fail in a task, I
keep trying until I succeed*

**Latent Space
(Unobservable Variables)**

Conscientiousness

ITEM_1

ITEM_2

ITEM_3

ITEM_4

**Manifest Space
(Observed Variables)**

The need for Latent Variable Models

Reliability: Consistency (precision) of a measure

- Reliability (General formula)

$\rho_{xx} = [Var(T) / Var(T + E)]$, with values ranging from 0 to 1

- Cronbach's Alpha

$$X = Y_1 + Y_2 + Y_3 + \dots + Y_k$$

$$\alpha = \frac{K}{K-1} \left(1 - \frac{\sum_{i=1}^K \sigma_{Y_i}^2}{\sigma_X^2} \right)$$

K = number of components (e.g., items)
 $\sigma_{Y_i}^2$ = variance of the component i
 σ_X^2 = variance of the observed total score

- Omega total

$$\omega_{Total} = \frac{\left(\sum_{i=1}^k \lambda_i \right)^2}{\left(\sum_{i=1}^k \lambda_i \right)^2 + \sum_{i=1}^k \theta_{ii}}$$

λ_i = factor loading of the component i ;
 a factor loading represents the strenght of the relationship between an indicator and its latent variable

θ_{ii} = Residual variance of the component i

- AVE (Average Variance Extracted) [Convergent Validity]

$$AVE = \frac{(\sum_{i=1}^k \lambda_i^2)}{(\sum_{i=1}^k \lambda_i^2) + (\sum_{i=1}^k \theta_{ii})} > .50$$

The need for Latent Variable Models

Validity: “A test is valid if it measures what it purports to measure” (Borsboom et al., 2004, p. 1061).

“[...] whether a test really measures what it purports to measure” (Kelley, 1927, p. 14)

“The problem of validity cannot be solved by psychometric techniques or models alone. On the contrary, it must be addressed by substantive theory”.

(Borsboom et al., 2004, p. 1062).

Psychological Review
2004, Vol. 111, No. 4, 1061–1071

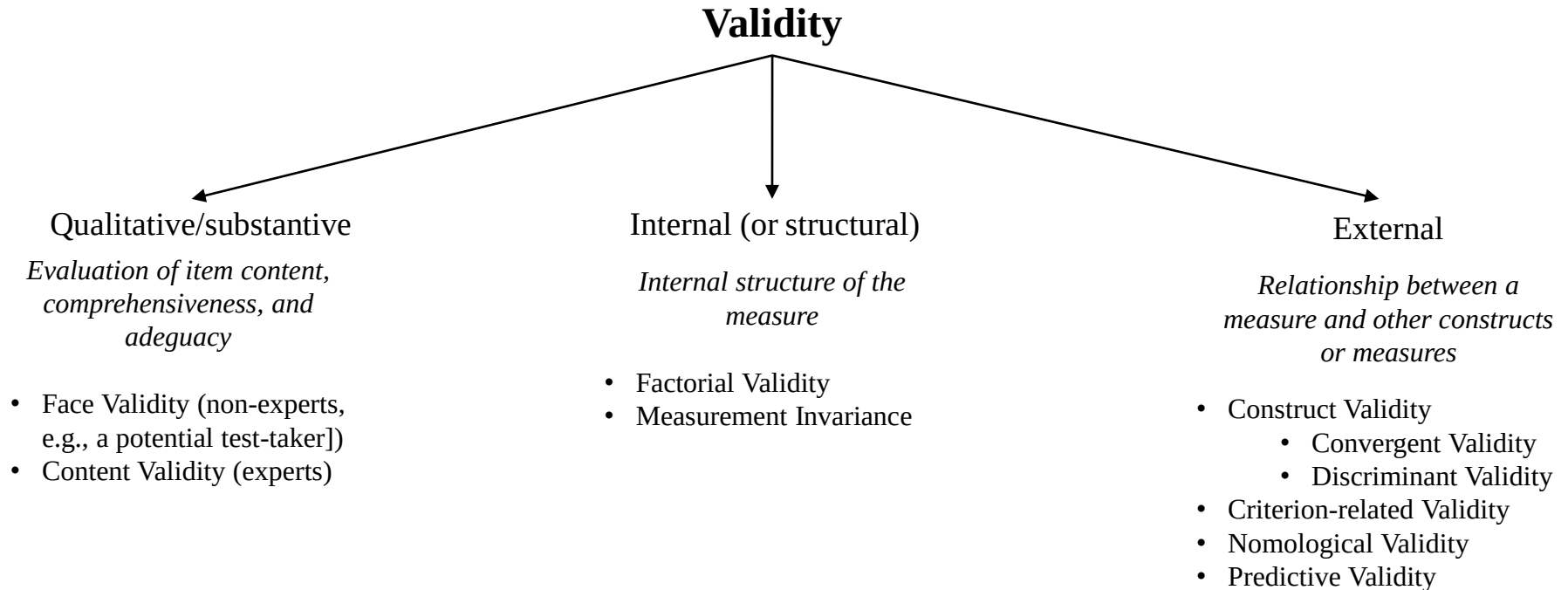
Copyright 2004 by the American Psychological Association
0033-295X/04/\$12.00 DOI: 10.1037/0033-295X.111.4.1061

The Concept of Validity

Denny Borsboom and Gideon J. Mellenbergh
University of Amsterdam

Jaap van Heerden
Maastricht University

The need for Latent Variable Models



The need for Latent Variable Models

Some references about validity

Some references to deepen Validity (particular focus on Work and Organiz Psych):

- Substantive Validity
 - Face Validity ([Allen et al., 2023](#))
 - Content Validity ([Colquitt et al., 2019](#); [Rossiter, 2008](#))
- Structural (Internal) Validity and Reliability
 - Reliability ([Cortina et al., 2020](#))
 - Measurement Invariance ([Somaraju et al., 2022](#))
- «External» Validity
 - Construct/Nomological/Criterion Validity ([Lambert & Newman, 2023](#))
 - Convergent Validity ([Cheung et al., 2024](#))
 - Discriminant Validity ([Cheung et al., 2024](#); [Rönkkö & Cho, 2022](#))

Overview typology of Validity ([Lim, 2024](#))



The need for Latent Variable Models

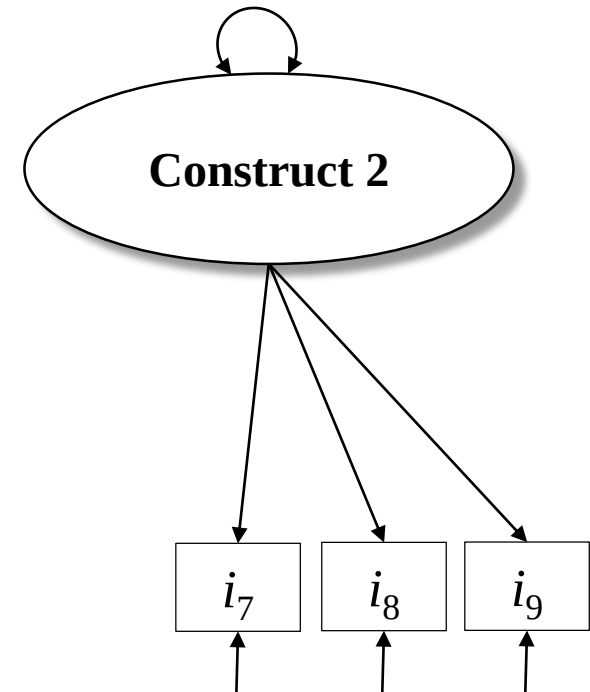
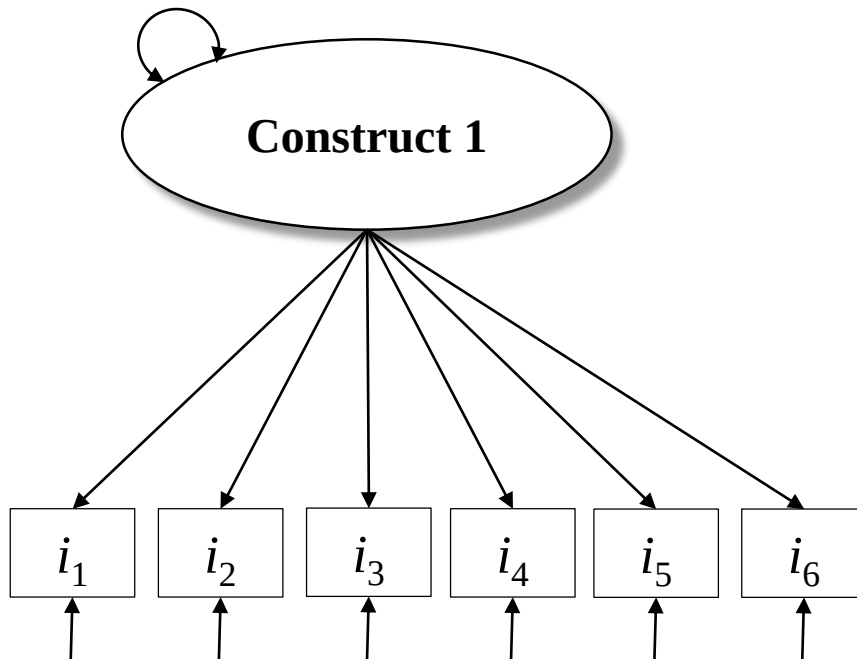
Researcher's main tasks in Personality Psychology

Measuring constructs
(Exploratory Factorial Analysis; EFA)

Connecting constructs
(Regression Models)

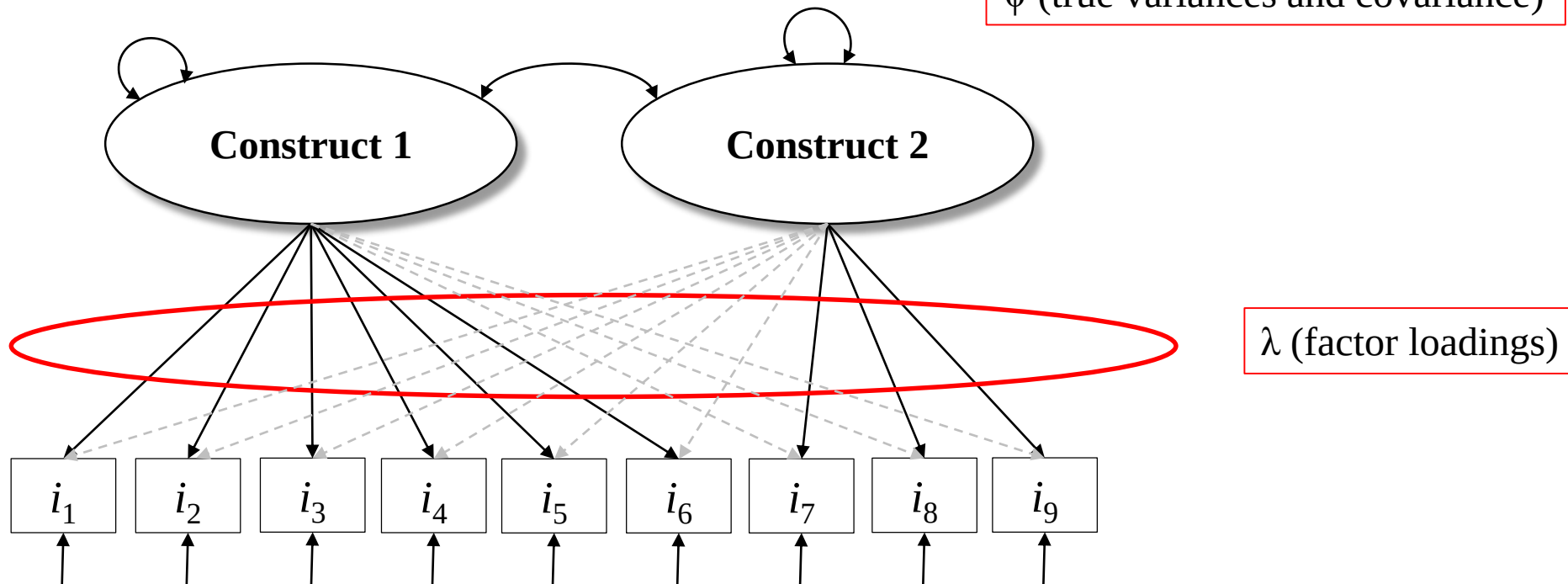
The need for Latent Variable Models

Measuring Constructs



The need for Latent Variable Models

Measuring Constructs (EFA)





Seeking Resources

1. I ask other feedback on my performance
2. I ask colleagues for advice [I ask other educational superintendents for advice]
3. I ask my supervisor for advice [I ask other managers for advice (not educational superintendents)]
4. I try to learn new things at work
5. I contacted other people from work (e.g., colleagues, supervisors) to get the necessary information for completing my tasks
6. When I have difficulties or problems at my work, I discuss them with people from my work environment

Seeking Challenges

7. I ask for more tasks if I finish my work [I look for other projects to carry out, as soon as I conclude those planned]
8. I ask for more responsibilities
9. I ask for more odd jobs [I try to engage in projects/assignments that test myself]

The Likert scale ranged from 1 = *never* to 5 = *often*.

Hypothesized Construct	Item	1	2
Seeking Resources ($\alpha = .71$)	1	.34***	.19***
	2	.59***	-.10***
	3	.53***	.18***
	4	.46***	.13***
	5	.38***	-.09***
	6	.76***	-.06**
Seeking Challenges ($\alpha = .83$)	7	-.05*	.61***
	8	.09**	.79***
	9	.02n.s.	.82***

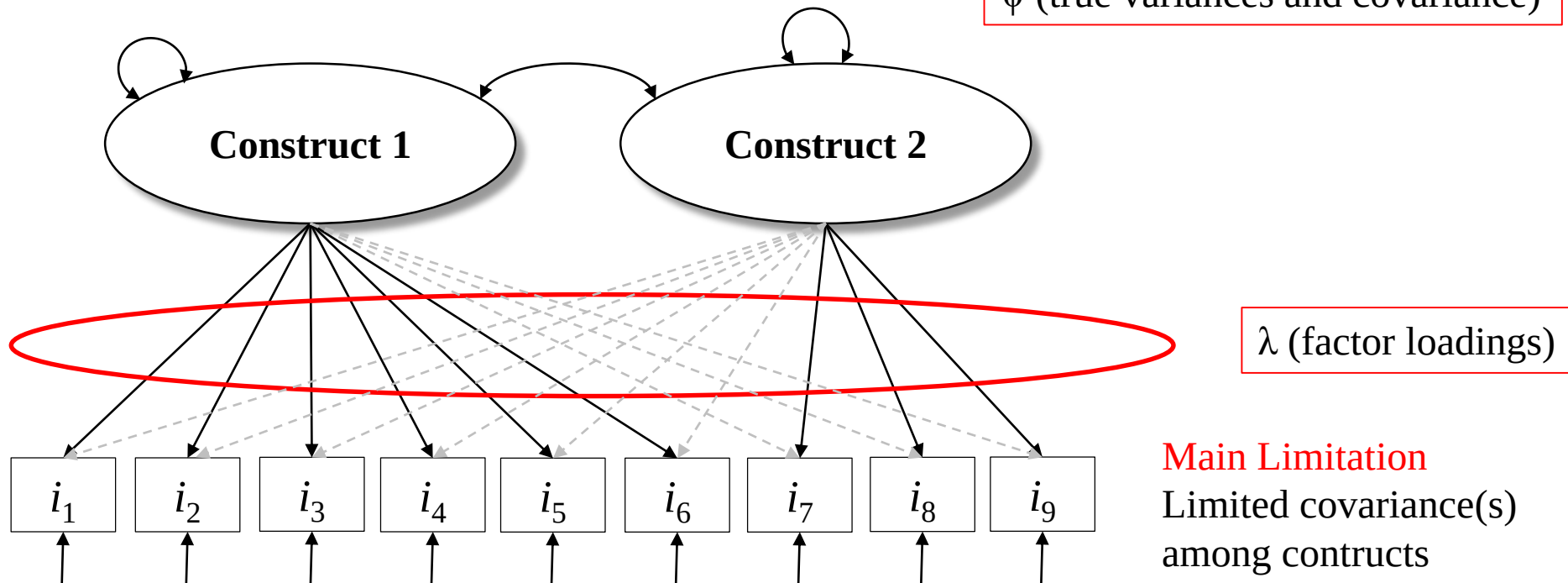
$N = 1060$ superintendents
Latent Correlation = .40***

Recommended thresholds:

- $|\text{Target } \lambda| > .30$
- $|\text{Non-target } \lambda| < .30$

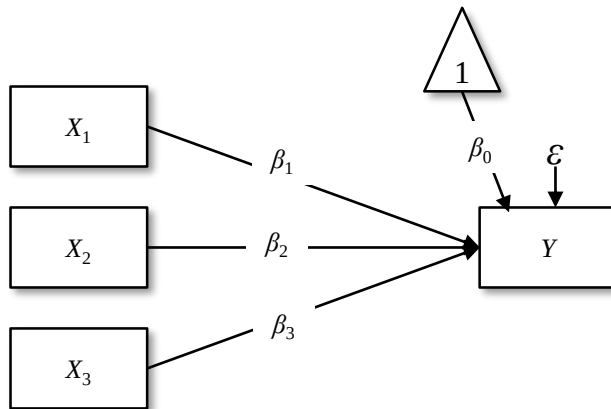
The need for Latent Variable Models

Measuring Constructs (EFA)



The need for Latent Variable Models

Connecting Constructs (Regression)



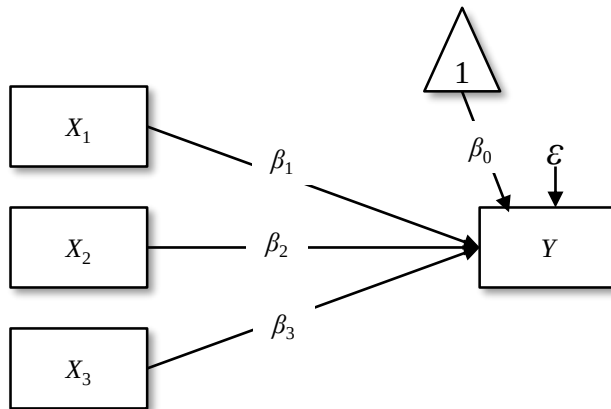
$$\min_{\mathbf{b}} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$
$$\hat{\mathbf{b}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$

} Basic of OLS

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \beta_3 X_{3i} + \varepsilon_i$$

The need for Latent Variable Models

Connecting Constructs (Regression)



$$\min_{\mathbf{b}} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$$\hat{\mathbf{b}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

} Basic of OLS

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \beta_3 X_{3i} + \varepsilon_i$$

Main Limitations

- 1) Don't take into account measurement error
- 2) One Dependent Variable

Structural Equation Modeling (SEM)

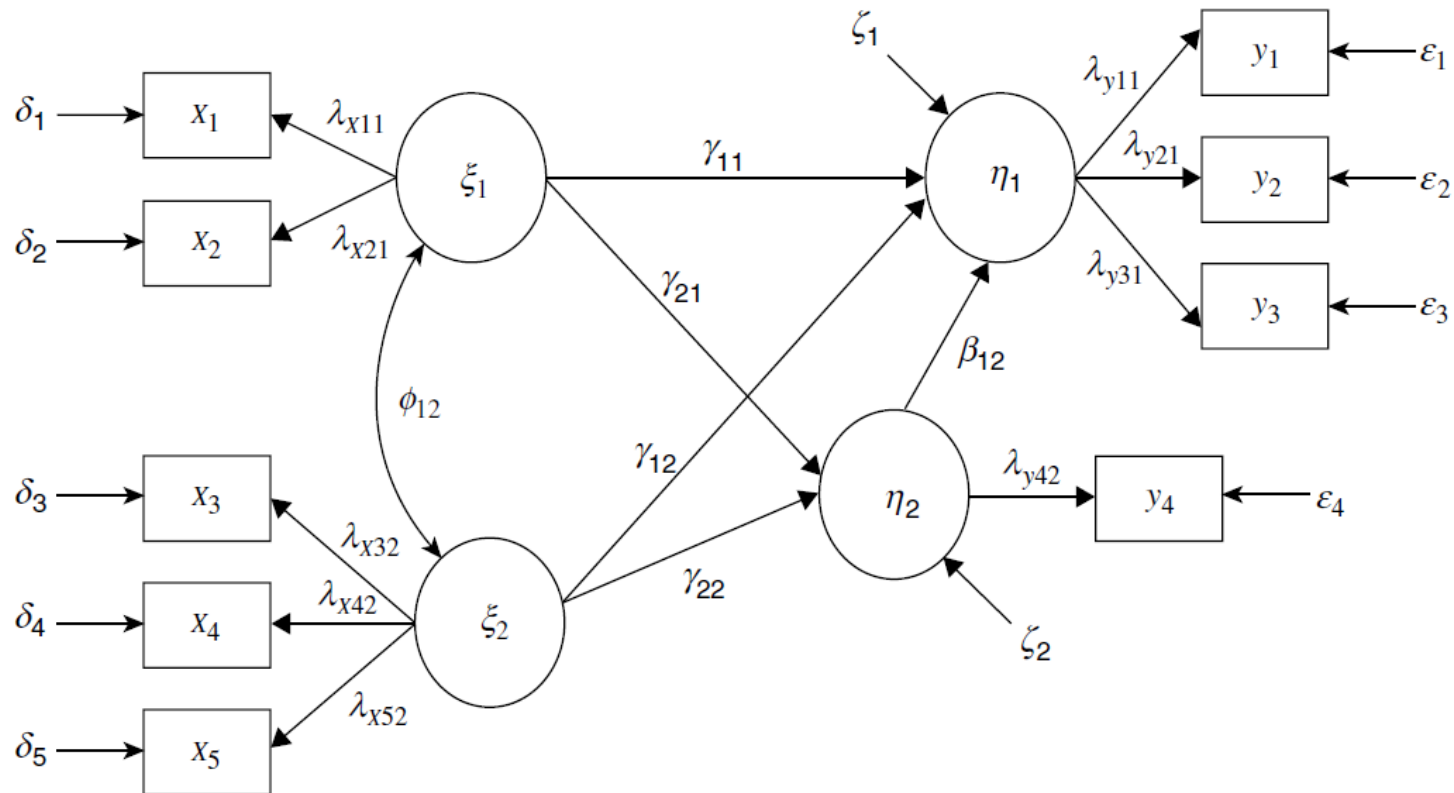
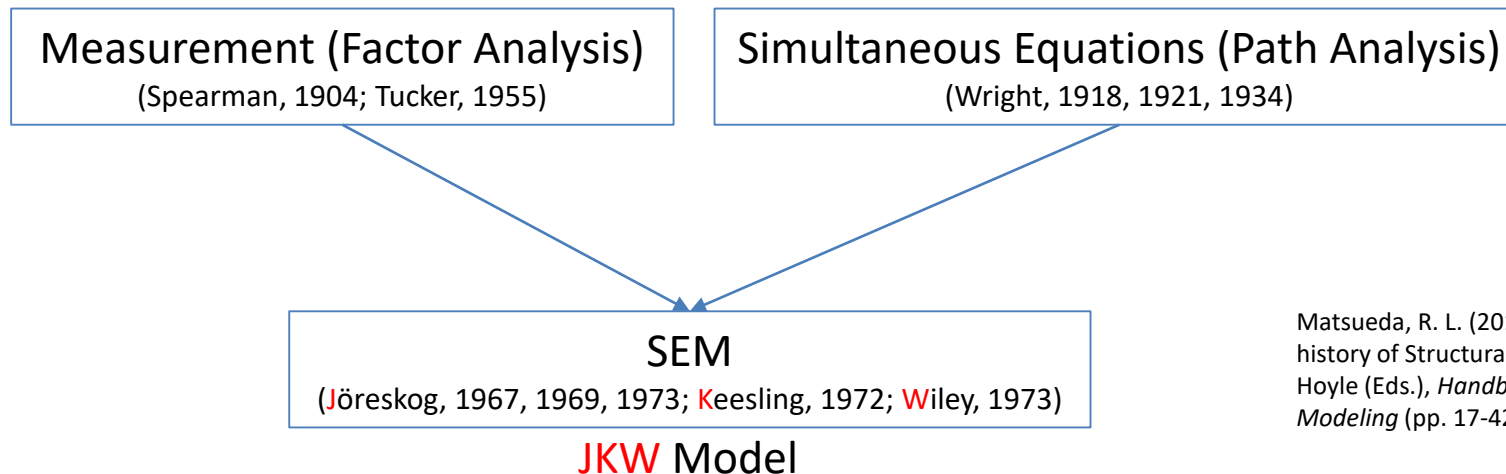


Figure 1.1 A hypothesized general structural equation model.

History of SEM

SEM (also called **covariance structure analysis**, **covariance structure modeling**, or **analysis of covariance structures**) does not designate a single statistical technique but instead refers to a family of related procedures (Kline, 2016, p. 9).

In SEM, unobservable latent variables (constructs or factors) are estimated from observed indicator variables, and the focus is on estimation of the relations among the latent variables free of the influence of measurement errors.



Matsueda, R. L. (2012). Key advances in the history of Structural Equation Modeling. In R. H. Hoyle (Eds.), *Handbook of Structural Equation Modeling* (pp. 17-42). New York, NY: Guilford

History of SEM

Matsueda (2012) argued that we can track **4 stages across SEM history**, each of which is characterized by a strong reliance on insights from statistics

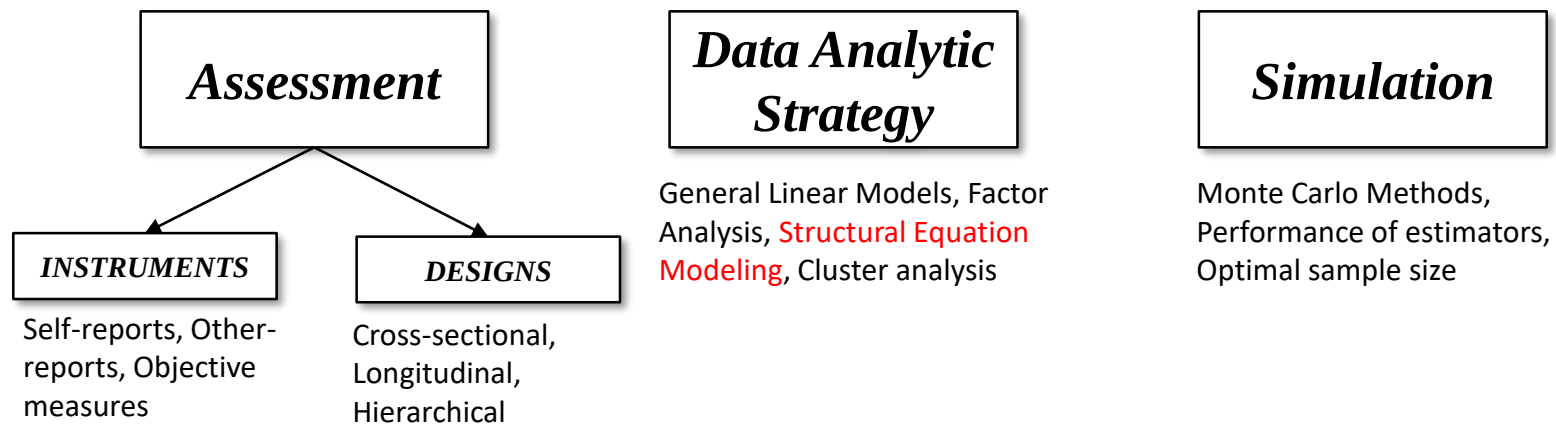
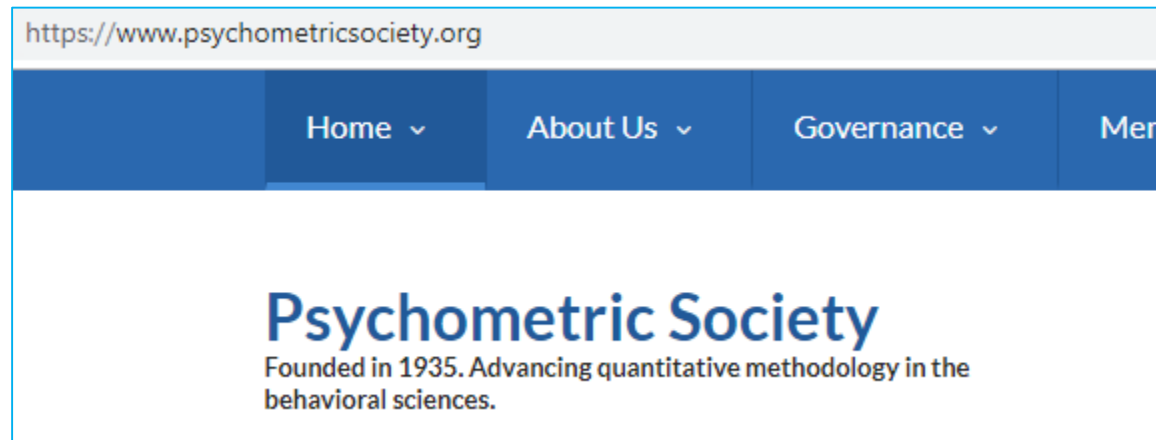
- (1) early disciplinary-specific developments of ***path analysis*** first from **genetics** and later **sociology**, ***factor analysis*** from **psychology**, and simultaneous equation models in **economics**;
- (2) cross-disciplinary fertilization between economics, sociology, and psychology, leading to an **explosion of empirical applications** of SEM;
- (3) a period of developing methods for handling **discrete, ordinal, and limited dependent variables**;
- (4) a recent period of incorporating **statistical advances** into the SEM framework, including generalized linear models, mixed effects models, **mixture** regression models, **Bayesian** methods, graphical models, and methods for identifying **causal effects**.

We are now approaching a fifth stage (e.g., Jacobucci & Grimm, 2020):

- (5) Latent variables ↔ data science, artificial intelligence (machine learning)

SEM and Psychometrics

The study of quantitative measurement in psychology



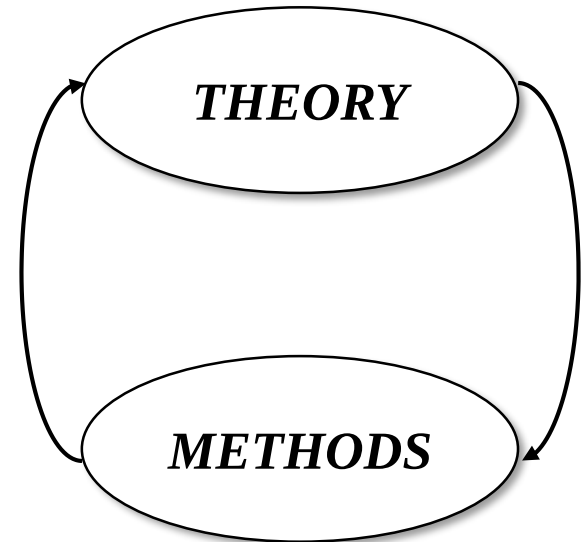
PSYCHOMETRIKA—VOL. 71, NO. 3, 425–440
SEPTEMBER 2006
DOI: 10.1007/s11336-006-1447-6

THE ATTACK OF THE PSYCHOMETRICIANS

DENNY BORSBOOM

UNIVERSITY OF AMSTERDAM

This paper analyzes the theoretical, pragmatic, and substantive factors that have hampered the integration between psychology and psychometrics. Theoretical factors include the operationalist mode of thinking which is common throughout psychology, the dominance of classical test theory, and the use of “construct validity” as a catch-all category for a range of challenging psychometric problems. Pragmatic factors include the lack of interest in mathematically precise thinking in psychology, inadequate representation of psychometric modeling in major statistics programs, and insufficient mathematical training in the psychological curriculum. Substantive factors relate to the absence of psychological theories that are sufficiently strong to motivate the structure of psychometric models. Following the identification of these problems, a number of promising recent developments are discussed, and suggestions are made to further the integration of psychology and psychometrics.



“The founding fathers of the Psychometric Society—scholars such as Thurstone, Thorndike, Guilford, and Kelley—**were substantive psychologists as much as they were psychometricians.**

Contemporary psychometricians do not always display a comparable interest with respect to the substantive field that lends them their credibility.

It is perhaps worthwhile to emphasize that, even though psychometrics has benefited greatly from the input of mathematicians, **psychometrics is not a pure mathematical discipline but an applied one.** If one strips the application from an applied science one is not left with very much that is interesting; and **psychometrics without the “psycho” is not, in my view, an overly exciting discipline.** It is therefore **essential that a psychometrician keeps up to date with the developments in one or more subdisciplines of psychology.**” (Borsboom, 2006, p. 436)

Structural Equation Modeling

Steps

1. **Model formulation** Specifying the SEM model that the researcher wants to test. The model may be formulated on the basis of theory or empirical findings. A general SEM model is composed of two parts: the measurement model and the structural model.
2. **Model identification** It determines whether there is a unique solution for all the free parameters in the specified model.
3. **Model estimation** Estimate model parameters and generate fitting function. Various estimation methods are available for SEM (e.g., ML, WLS, Bayes, ecc.).
4. **Model evaluation** Assess whether the model fits the data.
5. **Model modification** If the model does not fit the data, re-specification or modification of the model is needed. In this instance, the researcher makes a decision regarding how to delete, add, or modify parameters in the model.

Model Formulation

Steps

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Model Formulation

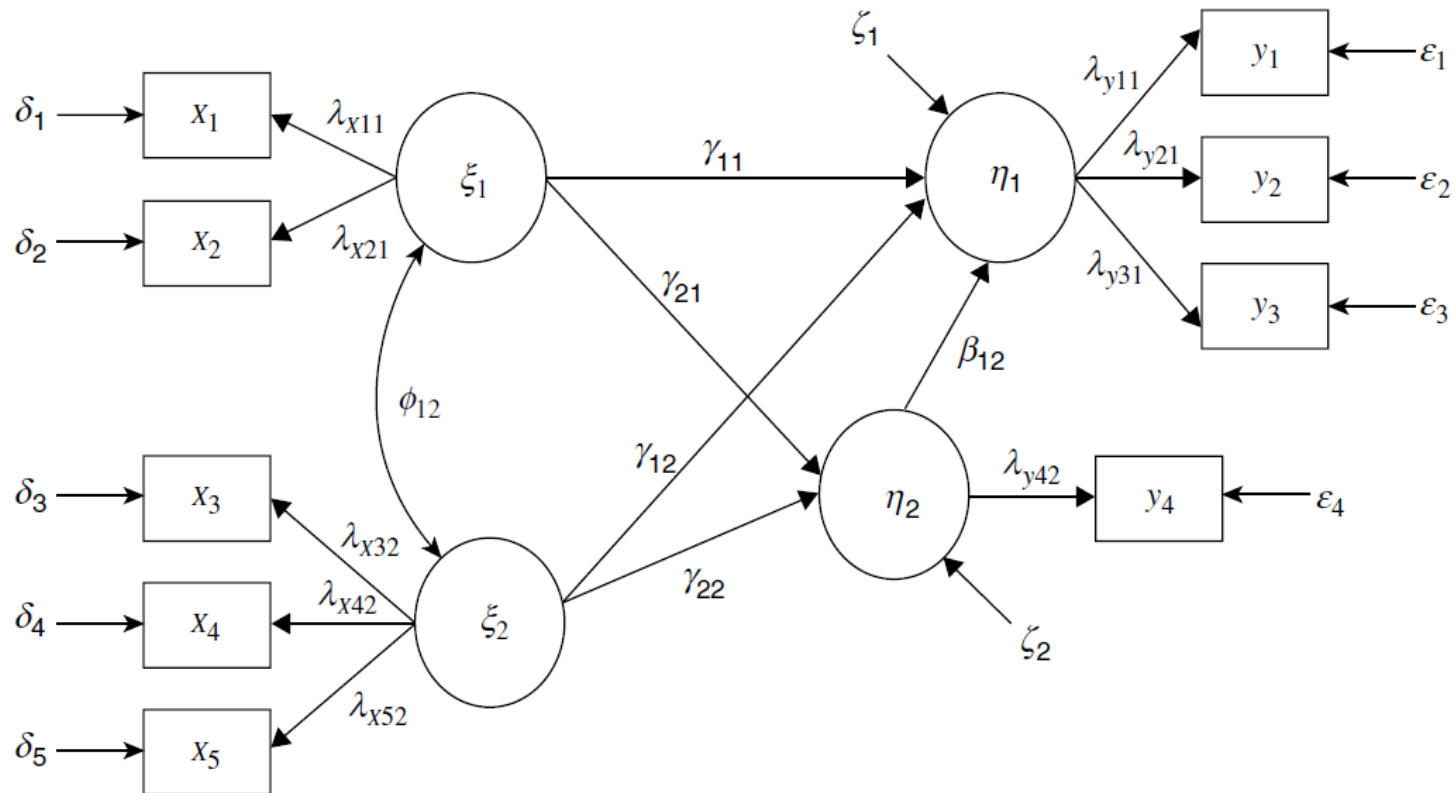


Figure 1.1 A hypothesized general structural equation model.

Model Formulation

Measurement Models

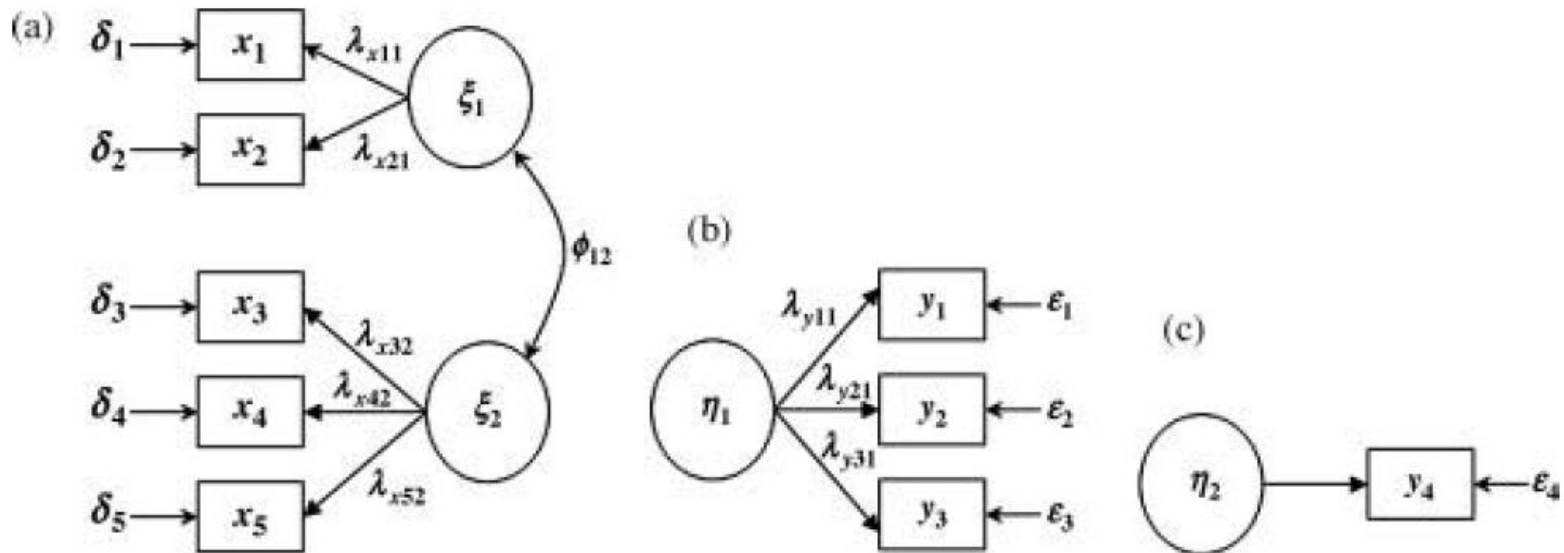
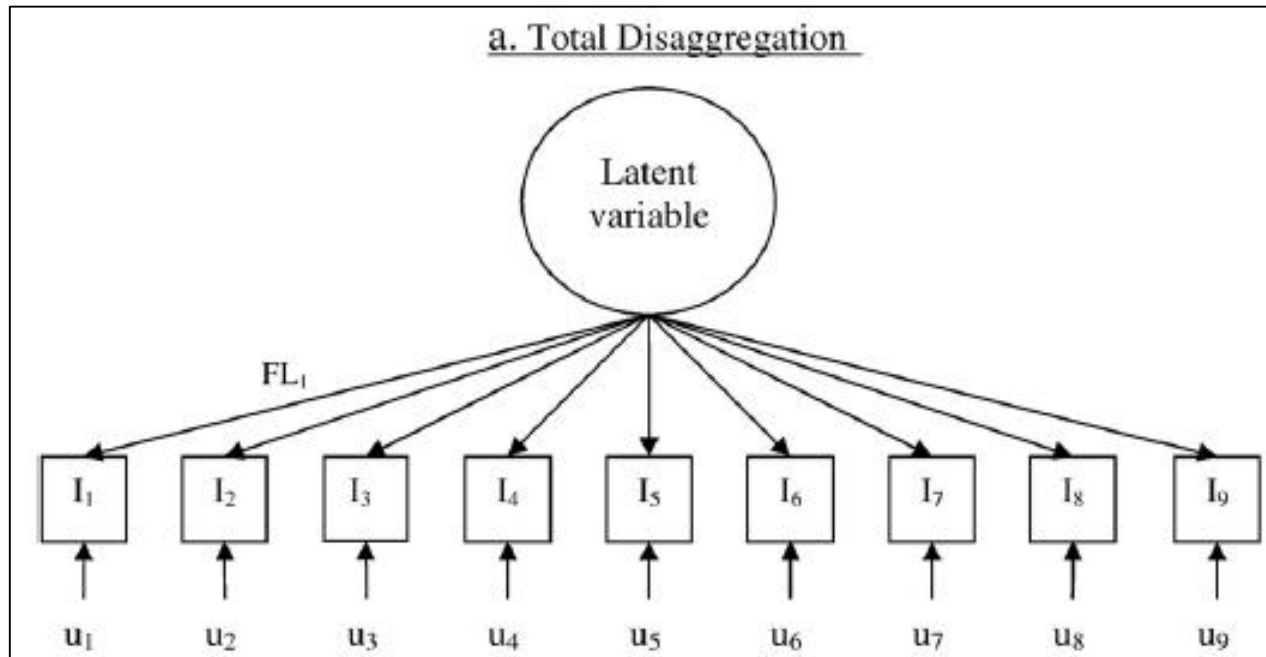


Figure 1.2 (a) Measurement model 1. (b) Measurement model 2. (c) Measurement model 3.

Model Formulation

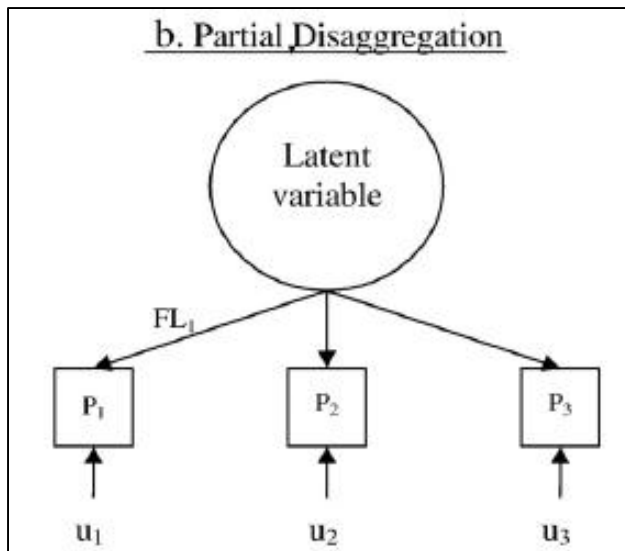
Types of Measurement Models



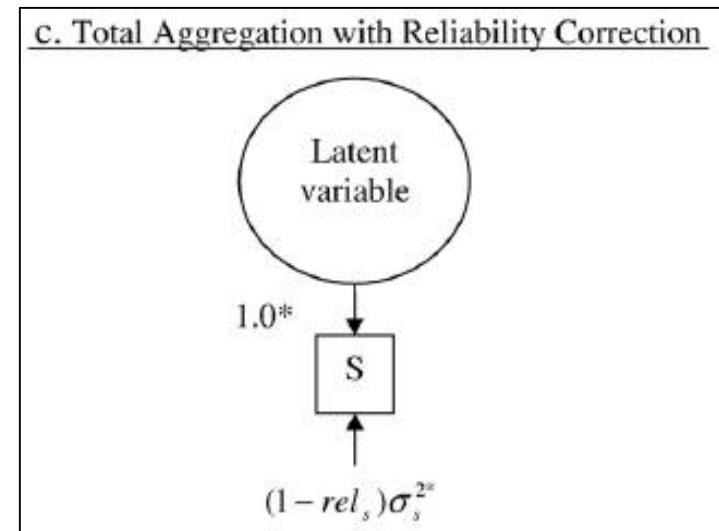
- I = Single items

Model Formulation

Types of Measurement Models



- $P = \text{Parcels}$
 - *Balancing technique*
 - *Facet-representative parceling*
 - *Domain-representative parceling*
- (Little, Rioux, Odejimi, & Stickley, 2022)



- $S = (\text{composite}) \text{ Score}$

Model Formulation

Structural Model

The term "structural" stands for the assumption that the parameters are not just descriptive measures of association but rather that they reveal an invariant "causal" relation.
(Bollen, 1989, p. 4)

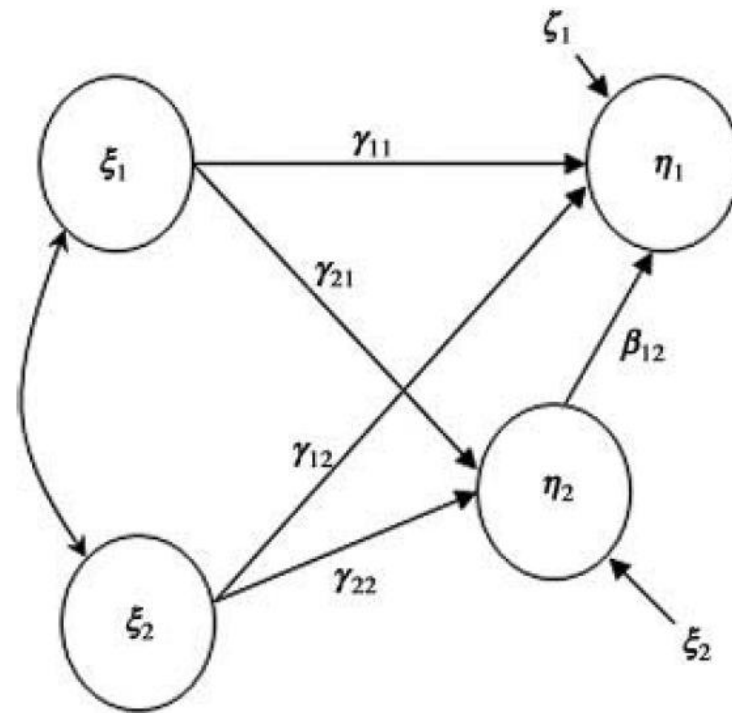


Figure 1.3 Structural model.



Model Formulation

Variables

Variable	Definition	Dimension
η (eta)	Latent endogenous variable	$m \times 1$
ξ (xi)	Latent exogenous variable	$n \times 1$
ζ (zeta)	Residual term in equations	$m \times 1$
y	Endogenous indicators	$p \times 1$
x	Exogenous indicators	$q \times 1$
ε (epsilon)	Measurement errors of y	$p \times 1$
δ (delta)	Measurement errors of x	$q \times 1$

exogenous = independent

endogenous = dependent (but can also be predictors!!)

n = latent exogenous variables (ξ)

m = latent endogenous variables (η)

q = exogenous indicators (x)

p = endogenous indicators (y)

Model Formulation

Matrices

Matrix	Definition	Dimension
Coefficient matrices		
Λ_y (lambda y)	Factor loadings relating y to η	$p \times m$
Λ_x (lambda x)	Factor loadings relating x to ξ	$q \times n$
B (beta)	Coefficient matrix relating η to η	$m \times m$
Γ (gamma)	Coefficient matrix relating ξ to η	$m \times n$
Variance/covariance matrices		
Φ (phi)	Variance/covariance matrices of ξ	$n \times n$
Ψ (psi)	Variance/covariance matrices of ζ	$m \times m$
Θ_ε (theta-epsilon)	Variance/covariance matrices of ε	$p \times p$
Θ_δ (theta-delta)	Variance/covariance matrices of δ	$q \times q$



Model Formulation

Equations

Structural Model $\eta = \mathbf{B}\eta + \mathbf{\Gamma}\xi + \zeta$

Measurement Models $\mathbf{Y} = \mathbf{\Lambda}_y\eta + \varepsilon$

$$\mathbf{X} = \mathbf{\Lambda}_x\xi + \delta$$

Model Formulation

Equations

Structural Model $\boldsymbol{\eta} = \mathbf{B}\boldsymbol{\eta} + \boldsymbol{\Gamma}\boldsymbol{\xi} + \boldsymbol{\zeta}$

Measurement Models $\mathbf{Y} = \boldsymbol{\Lambda}_y\boldsymbol{\eta} + \boldsymbol{\varepsilon}$

$$\mathbf{X} = \boldsymbol{\Lambda}_x\boldsymbol{\xi} + \boldsymbol{\delta}$$

$$\begin{bmatrix} \eta_1 \\ \eta_2 \end{bmatrix} = \begin{bmatrix} 0 & \beta_{12} \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \eta_1 \\ \eta_2 \end{bmatrix} + \begin{bmatrix} \gamma_{11} & \gamma_{12} \\ \gamma_{21} & \gamma_{22} \end{bmatrix} \begin{bmatrix} \xi_1 \\ \xi_2 \end{bmatrix} + \begin{bmatrix} \zeta_1 \\ \zeta_2 \end{bmatrix}$$

$$\eta_{1i} = \beta_{12} \eta_{2i} + \gamma_{11} \xi_{1i} + \gamma_{12} \xi_{2i} + \zeta_{1i}$$

$$\eta_{2i} = \gamma_{21} \xi_{1i} + \gamma_{22} \xi_{2i} + \zeta_{2i}$$

Model Formulation

Equations

Structural Model

$$\eta = B\eta + \Gamma\xi + \zeta$$

Measurement Models

$$Y = \Lambda_y \eta + \varepsilon$$

$$X = \Lambda_x \xi + \delta$$

$$\begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ \lambda_{y21} & 0 \\ \lambda_{y31} & 0 \\ 0 & \lambda_{y42} \end{bmatrix} \begin{bmatrix} \eta_1 \\ \eta_2 \end{bmatrix} + \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \\ \varepsilon_4 \end{bmatrix}$$

$$y_{1i} = \eta_{1i} + \varepsilon_{1i}$$

$$y_{2i} = \lambda_{y21} \eta_{1i} + \varepsilon_{2i}$$

$$y_{3i} = \lambda_{y31} \eta_{1i} + \varepsilon_{3i}$$

$$y_{4i} = \lambda_{y42} \eta_{2i} + \varepsilon_{4i}$$

Model Formulation

Equations

Structural Model

$$\eta = B\eta + \Gamma\xi + \zeta$$

Measurement Models

$$Y = \Lambda_y\eta + \varepsilon$$

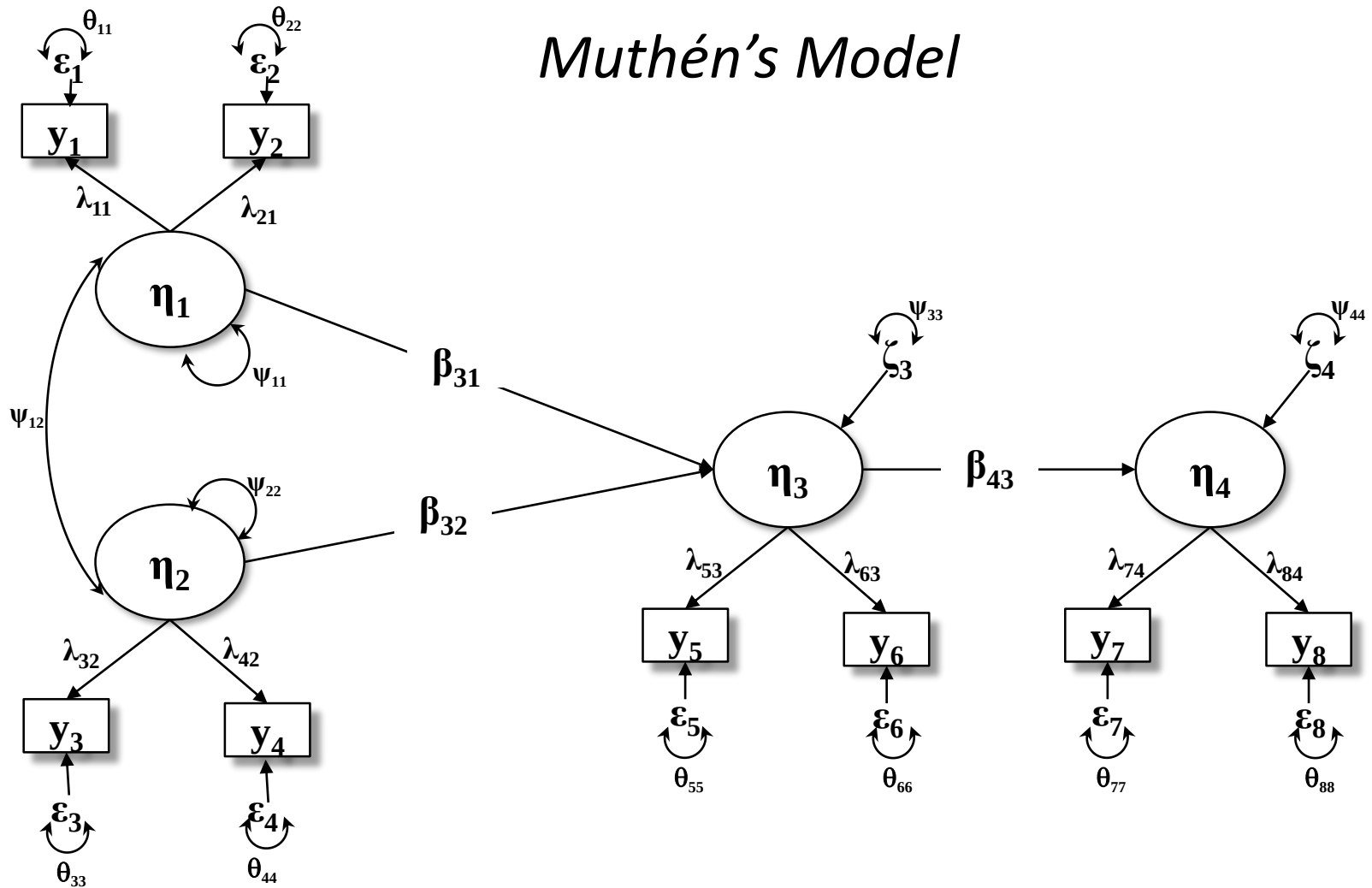
$$X = \Lambda_x\xi + \delta$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ \lambda_{x21} & 0 \\ 0 & 1 \\ 0 & \lambda_{x42} \\ 0 & \lambda_{x52} \end{bmatrix} \begin{bmatrix} \xi_1 \\ \xi_2 \end{bmatrix} + \begin{bmatrix} \delta_1 \\ \delta_2 \\ \delta_3 \\ \delta_4 \\ \delta_5 \end{bmatrix}$$

$$\begin{aligned} x_{1i} &= \xi_{1i} + \delta_{1i} \\ x_{2i} &= \lambda_{x21} \xi_{1i} + \delta_{2i} \\ x_{3i} &= \xi_{1i} + \delta_{3i} \\ x_{4i} &= \lambda_{x42} \xi_{2i} + \delta_{4i} \\ x_{5i} &= \lambda_{x52} \xi_{2i} + \delta_{5i} \end{aligned}$$

Model Formulation

Muthén's Model



Model Formulation

Muthén's Model

Structural Model $\eta = B\eta + F\xi + \zeta$

Measurement Models $Y = \Lambda_y \eta + \varepsilon$

~~$X = \Lambda_x \xi + \delta$~~

Structural Model $\eta = B\eta + \zeta$

Measurement Model $Y = \Lambda\eta + \varepsilon$

By adding intercepts/means

Structural Model $\eta = \mathbf{v} + B\eta + \zeta$

Measurement Models $Y = \alpha + \Lambda\eta + \varepsilon$



Model Formulation

Exercise

- Draw a SEM or find it in a publication
- Specify which are the measurement models
- Specify the “type/s” of measurement model (i.e., total disaggregation, partial disaggregation, total aggregation with reliability correction)
- Specify what is the structural model

Model Identification

Steps

1. **Model formulation** Specifying the SEM model that the researcher wants to test. The model may be formulated on the basis of theory or empirical findings. A general SEM model is composed of two parts: the measurement model and the structural model.
2. **Model identification** It determines whether there is a unique solution for all the free parameters in the specified model.
3. **Model estimation** Estimate model parameters and generate fitting function. Various estimation methods are available for SEM (e.g., ML, WLS, Bayes, ecc.).
4. **Model evaluation** Assess whether the model fits the data.
5. **Model modification** If the model does not fit the data, re-specification or modification of the model is needed. In this instance, the researcher makes a decision regarding how to delete, add, or modify parameters in the model.

Model Identification

Parameters

Model identification concerns whether a unique value for each and every unknown parameter can be estimated from the observed data (Wang & Wang, 2012, p. 11).

Parameters can be:

1. **FIXED:** forced to assume a specific value, such as 0 or 1
2. **CONSTRAINED:** to be equal to another parameter
3. **FREE:** estimated by the model

Model Identification

Degrees of Freedom

$$df = [p(p + 1)]/2 - \text{NFP} > 0$$

p = number of observed variables, NFP = Number of free parameters

	1.	2.	3.	4.	5.	6.	7.	8.
1. y_1	Var(y_1)							
2. y_2	Cov($y_1 y_2$)	Var(y_2)						
3. y_3	Cov($y_1 y_3$)	Cov($y_2 y_3$)	Var(y_3)					
4. y_4	Cov($y_1 y_4$)	Cov($y_2 y_4$)	Cov($y_3 y_4$)	Var(y_4)				
5. y_5	Cov($y_1 y_5$)	Cov($y_2 y_5$)	Cov($y_3 y_5$)	Cov($y_4 y_5$)	Var(y_5)			
6. y_6	Cov($y_1 y_6$)	Cov($y_2 y_6$)	Cov($y_3 y_6$)	Cov($y_4 y_6$)	Cov($y_5 y_6$)	Var(y_6)		
7. y_7	Cov($y_1 y_7$)	Cov($y_2 y_7$)	Cov($y_3 y_7$)	Cov($y_4 y_7$)	Cov($y_5 y_7$)	Cov($y_6 y_7$)	Var(y_7)	
8. y_8	Cov($y_1 y_8$)	Cov($y_2 y_8$)	Cov($y_3 y_8$)	Cov($y_4 y_8$)	Cov($y_5 y_8$)	Cov($y_6 y_8$)	Cov($y_7 y_8$)	Var(y_8)

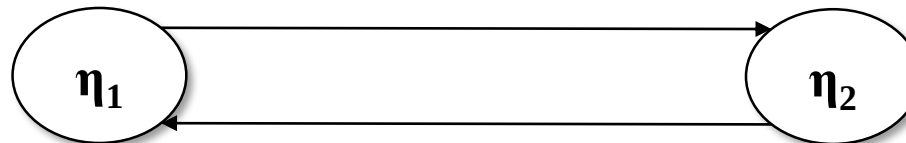
e.g. with 8 observed variables (p) we have $(8*9)/2 = 36$ non-redundant elements.
Thus, NFP could be up to 35.

Model Identification

Other assumptions

There are a number of other assumptions for ensuring identification (see Bollen, 1989), but the most important is that the model must be **recursive**.

An example of **nonrecursive** model is a model with loops (Rigdon, 1995):





Model Identification

Exercise

On your previous SEM (or another one)

- Compute the maximum number of possible free parameters
- Identify some fixed and free parameters (and constrained, if there are)

Model Estimation

Steps

1. **Model formulation** Specifying the SEM model that the researcher wants to test. The model may be formulated on the basis of theory or empirical findings. A general SEM model is composed of two parts: the measurement model and the structural model.
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Model Estimation

Estimation of SEM models is different from that of multiple regressions.

Instead of minimizing the discrepancies between the fitted and observed values of the response variable [i.e., $\Sigma(y - \hat{y})$], SEM estimation procedures minimize the residuals that are differences between the **sample variances/covariances** and the **variances/covariances** estimated from the model. (Wang & Wang, 2012, p. 14).

$$\Sigma = \Sigma(\theta)$$

Given that the **population variances/covariances matrix (Σ)** is unknown, it is replaced by the **sample variances/covariances matrix (S)**, so that the null hypothesis is

$$S = \Sigma(\theta)$$

Model Estimation

There are several estimators, but the most common procedure is the use of **Maximum Likelihood function**¹

$$F_{\text{ML}} = \log|\Sigma(\theta)| + \text{tr}(\mathbf{S}\Sigma^{-1}(\theta)) - \log|\mathbf{S}| - (p + q) \quad (4.67)$$

(Bollen, 1989, p. 107)

$(N - 1)F_{\text{ML}}$ is chi-squared distributed with $df = \{[p(p + 1)]/2 - \text{NFP}\}$, where N is the sample size and F_{ML} is the value of the fitting function evaluated at the final estimates (Bollen, 1989, p. 110).

Recall that the null hypothesis of this chi-square test is $H_0: \mathbf{S} = \Sigma(\theta)$.

Thus, a good model should show a **non-significant chi-square test**, because we expect **non-significant differences** between $\mathbf{S}$ and $\Sigma(\theta)$

Model Estimation

$$\log L_0 = -\frac{N-1}{2} \{ \log |\hat{\Sigma}| + \text{tr}(\hat{\Sigma}^{-1} \mathbf{S}) \} \quad (7.43)$$

$$\begin{aligned} \log L_1 &= -\frac{(N-1)}{2} \{ \log |\mathbf{S}| + \text{tr}(\mathbf{S}^{-1} \mathbf{S}) \} \\ &= -\frac{(N-1)}{2} \{ \log |\mathbf{S}| + q \} \end{aligned} \quad (7.47)$$

$$-2 \log \left(\frac{L_0}{L_1} \right) = -2 \log L_0 + 2 \log L_1$$

Model Estimation

Example from an Mplus Output

```
THE MODEL ESTIMATION TERMINATED NORMALLY

MODEL FIT INFORMATION

Number of Free Parameters          116

Loglikelihood

    H0 Value                      -5249.015
    H1 Value                      -4935.988

Information Criteria

    Akaike (AIC)                  10730.030
    Bayesian (BIC)                11167.155
    Sample-Size Adjusted BIC      10799.223
        (n* = (n + 2) / 24)

Chi-Square Test of Model Fit

    Value                         626.054
    Degrees of Freedom            397
    P-Value                       0.0000
```

Log-Likelihood for my specified model ($\log L_0$)

Log-Likelihood for the saturated model ($\log L_1$),
namely for the model in which $S = \Sigma(\theta)$

```
> (-2*-5249.015)+(2*(-4935.988))
[1] 626.054
```



Model Estimation

Exercise

Compute both chi-square statistics and degrees of freedom using the following information. **Pay attention!** Add the vector of means (for simplicity, again *p*) to the computation of degrees of freedom, that is $\{ \{ [p * (p + 1)] / 2 \} + p \} - \text{NFP}$

Number of observed variables	9
Number of Free Parameters	41

Loglikelihood

H0 Value	-1406.522
H1 Value	-1397.180



Model Estimation

Exercise

Compute both chi-square statistics and degrees of freedom using the following information. **Pay attention!** Add the vector of means (for simplicity, again p) to the computation of degrees of freedom, that is $\{ \{ [p*(p+1)]/2 \} + p \} - \text{NFP}$

Number of observed variables	9
Number of Free Parameters	41

Loglikelihood

H0 Value	-1406.522
H1 Value	-1397.180

Solution

Chi-square = $(-2 * -1406.522) + (2 * -1397.180) = \mathbf{18.684}$

df = $((9 * 10) / 2) + 9 - 41 = \mathbf{13}$

Model Evaluation

Steps

1. **Model formulation** Specifying the SEM model that the researcher wants to test. The model may be formulated on the basis of theory or empirical findings. A general SEM model is composed of two parts: the measurement model and the structural model.
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Model Evaluation

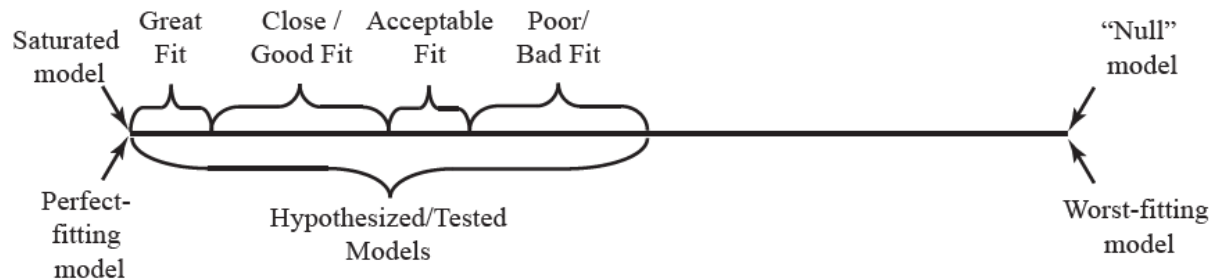


FIGURE 5.1. Continuum of conceptual relations among models. *Note.* This continuum goes from the worst-fitting, “null” model on the right to the perfect-fitting, saturated model on the left, with the hypothesized model and the different categories of fit in between. These category labels are loose terms, and the precise division between categories varies.

Null model. A model that attempts to gauge the amount of information in a sample of data. The null model is the worst-fitting model that can be specified while being theoretically defensible. The default null model is often termed the independence model. The independence model specifies no covariances among the observed indicators but does estimate the observed variances and means of the indicators.

Saturated model. A model in which there are just as many estimates as there are observed bits of information (and all parameters are locally identified). That is, such a model would have no degrees of freedom and would perfectly reproduce the matrix S and the observed means.

Model Evaluation

Absolute fit indices. Indices that compare the hypothesized model to the saturated model and therefore assess the degree of decrement from a perfectly fitting model.

- χ^2
 - RMSEA (Noncentrality-based Indices)
 - SRMR
 - AIC (Information Criteria)
 - BIC (Information Criteria)
- } The smaller the better

Relative fit indices. Relative measures of model fit rely on a null model to gauge model fit. Because the null model provides a measure of bad model fit (in fact, the worst-fitting model), it can be used to index the relative improvement in model fit that a hypothesized model provides.

- CFI (Noncentrality-based Indices)
 - TLI
- } The higher the better

Little (2024, p. 168 and 169)

See also Newsom

https://web.pdx.edu/~newsomj/semclass/ho_fit.pdf

Model Evaluation

χ^2 test is highly sensitive to sample size and to the number of variables included in the model.

“As such, the significance of the χ^2 test should not be a reason by itself to reject a model” (Wang & Wang, 2012, p. 18).

Model Evaluation

Other Common Indices of Fit

$$CFI = \frac{d_{null} - d_{specified}}{d_{null}} > .90-.95$$

d = noncentrality parameter = $\chi^2 - df$
 $null$ = the worst model (all cov. are zero)

$$TLI = \frac{\left(\frac{\chi_{null}^2}{df_{null}} - \frac{\chi_{specified}^2}{df_{specified}} \right)}{\left(\frac{\chi_{null}^2}{df_{null}} - 1 \right)} > .90-.95$$

$$RMSEA = \sqrt{\frac{\chi^2 - df}{(n-1)df}} < .08 - .05$$

$$SRMR = \sqrt{\frac{\sum_{i=1}^p \sum_{j=1}^i \left(\frac{s_{ij} - \hat{\sigma}_{ij}}{s_{ii}s_{jj}} \right)^2}{p(p+1)/2}} = \sqrt{\frac{\text{sum of squared std residuals}}{\text{size var-covar matrix}}} < .08 - .05$$

Model Evaluation

Nested models

Definition:

- Models “are *nested* [when] one model is produced by changing the status of one or more parameters in the other” (Hoyle, 2012, p. 11).
- “*Nested* models are those for which the parameter space associated with one model is a subset of the parameter space associated with the other model” (Merkle, You, & Preacher, 2016, p. 152-153)
- “Two models are nested if they are of the same form, but the free parameters of one model are a subset of the free parameters of the other model” (Hoyle, 2012, p. 141)

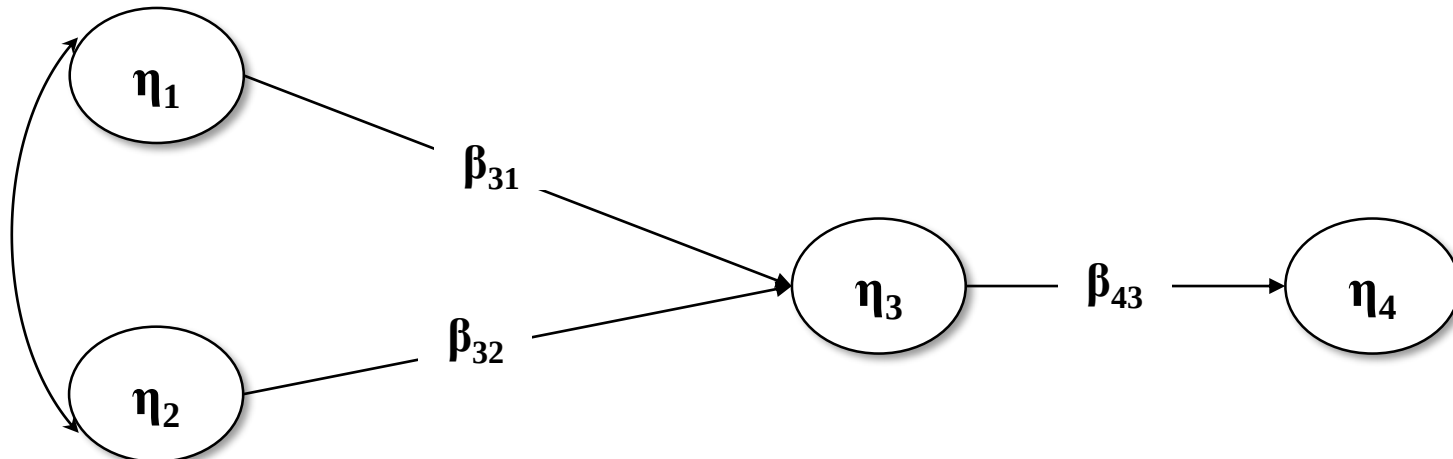
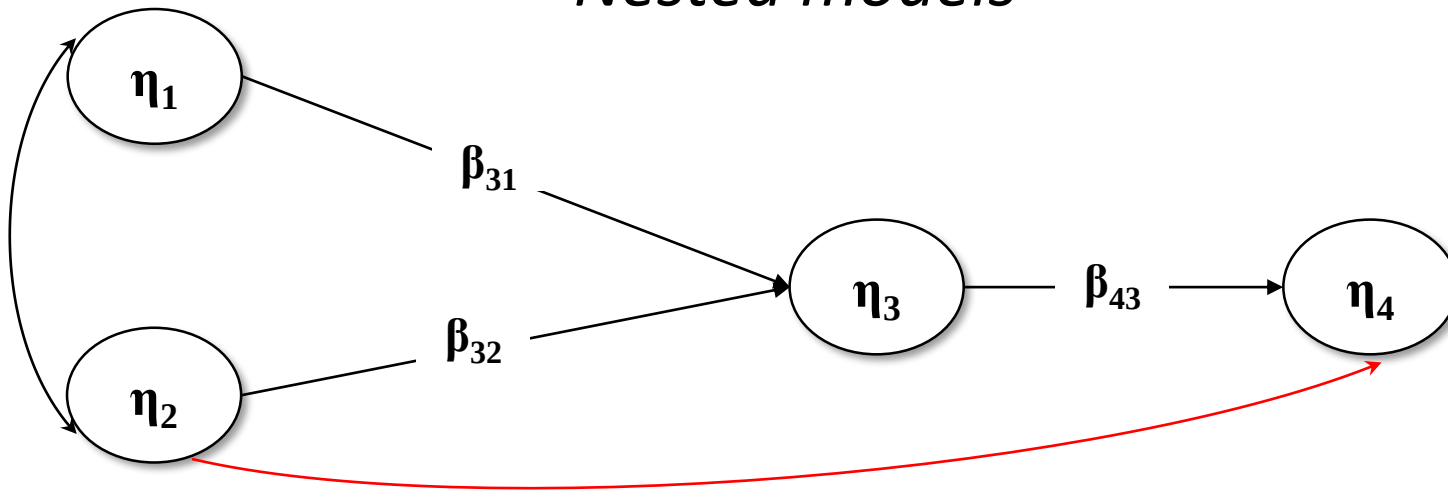
Nested models can be compared by means of $\Delta\chi^2$ statistics, distributed with Δdf .

The resulting *p*-value will attest

- whether removing 1 or more parameters significantly worsen (or not) the fit of the model
- whether adding 1 or more parameters significantly improve the fit of the model

Model Evaluation

Nested models



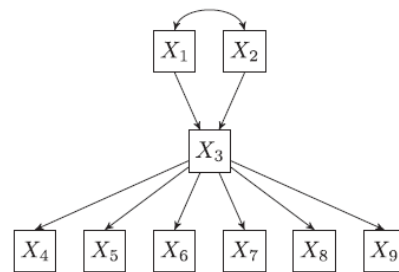
$\Delta df = 1$

Model Evaluation

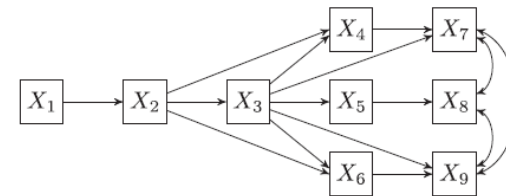
Non-Nested models

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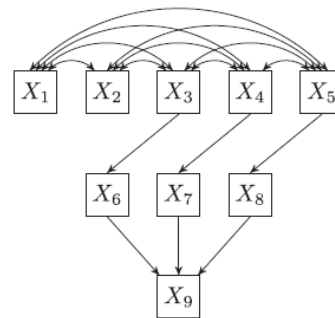
MERKLE, YOU, AND PREACHER



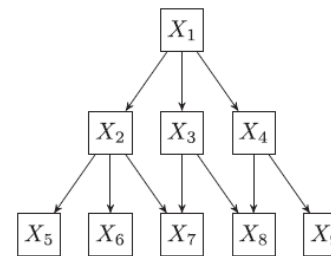
Model A ($df = 27$)



Model B ($df = 21$)



Model C ($df = 20$)



Model D ($df = 26$)

Figure 2. Path diagrams reflecting the models used in Simulation 2.

Model Evaluation

Information Criteria Indices

Useful for comparing **non-nested model**

$$AIC = -2\text{Ln}(L) + 2q$$

$$BIC = -2\text{Ln}(L) + q\text{Ln}(N)$$

$$ABIC = -2\text{Ln}(L) + q\text{Ln}[(N+2)/24]$$

$\text{Ln}(L)$ = log-likelihood value

q = number of free parameters

N = sample size

The model with the **smaller values** in these indices is preferred.

**IMPORTANT: YOU CANNOT USE THESE INDICES WHEN THE $\log L_1$ IS DIFFERENT
ACROSS MODELS**

Model Evaluation

Exercise: Fit and Nested Model Comparison

Table 2. Model fits of the predicted and alternative model tests.

	χ^2	df	<i>p</i>	CFI	TLI	RMSEA	SRMR	AIC	MC	$\Delta\chi^2$	Δdf	<i>p</i>
M1: No constraint	12.794	9	.172	0.991	0.967	0.066	0.022	2897.154				
M2: CS → OI fixed to be zero	13.819	11	.243	0.993	0.980	0.052	0.027	2894.179	M2 Vs M1	1.025	2	0.599
M3: PD → OI fixed to be zero	18.684	13	.133	0.986	0.966	0.067	0.041	2895.044	M3 Vs M2	4.865	2	0.088
M4 PD → CS fixed to be zero	29.094	15	.016	0.966	0.926	0.099	0.073	2901.454	M4 Vs M3	10.410	2	0.005

Note: CS = Colleague Support; OI = Organizational Identification, PD = Psychological Distress. CFI = Comparative Fit Index; TLI = Tucker-Lewis Index; RMSEA = Root Mean Square Error of Approximation; SRMR = Standardized Root Mean Square Residual; MC = Model Comparison. M1 = no constraint were imposed; M2 = the paths from colleague support T1 to organizational identification T2 and from colleague support T2 to organizational identification T3 were fixed to be zero; M3 = the paths from psychological distress T1 to organizational identification T2 and from psychological distress T2 to organizational identification T3 were fixed to be zero; M4 = the paths from psychological distress T1 to colleague support T2 and from psychological distress T2 to colleague support T3 were fixed to be zero.

Model Evaluation

Exercise: Fit and Non-Nested Model Comparison

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Table 3 Indices of hypothesized models (see Fig. 2) and alternative models ($N = 109$)

Model	Free Parameters	$\chi^2(df)$	CFI	TLI	RMSEA[90% CI]	SRMR	AIC
Hyp. mod.	21	10.10 (6) ^{n.s.}	.974	.934	.080[.000, .163]	.037	3085.58
Alt. mod. 1	23	6.55 (4) ^{n.s.}	.984	.939	.077[.000, .179]	.026	3086.03
Alt. mod. 2	22	8.98 (5) ^{n.s.}	.975	.924	.086[.000, .176]	.038	3086.47
Alt. mod. 3	21	12.9 (6) [*]	.955	.888	.104[.017, .183]	.048	3088.45

See paragraph “Alternative Models” in **Results** section for the description of each alternative model

All the models had the same maximum log-likelihood value for the unrestricted (H1) model, thus they are comparable with AIC

Hyp. mod. hypothesized model (see Fig. 2), *Alt. mod.* alternative model, *df* degrees of freedom, *CFI* Comparative Fit Index, *TLI* Tucker-Lewis Index, *RMSEA* Root Mean Square Error of Approximation, *CI* Confidence interval, *SRMR* Standardized Root Mean Square Residual, *AIC* Akaike Information Criterion

^{n.s.} $p > .10$, ^{*} $p < .05$

Model Modification

Steps

1. **Model formulation** Specifying the SEM model that the researcher wants to test. The model may be formulated on the basis of theory or empirical findings. A general SEM model is composed of two parts: the measurement model and the structural model.
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Model Modification

“If the model does not fit the data, re-specification or modification of the model is needed. In this instance, the researcher makes a decision regarding how to delete, add, or modify parameters in the model” (Wang & Wang, 2012, p. 2).

Modification Indices (MI): Change in χ^2 (i.e., $\Delta\chi^2$) after freeing a parameter

Usually, MI are provided in the output.

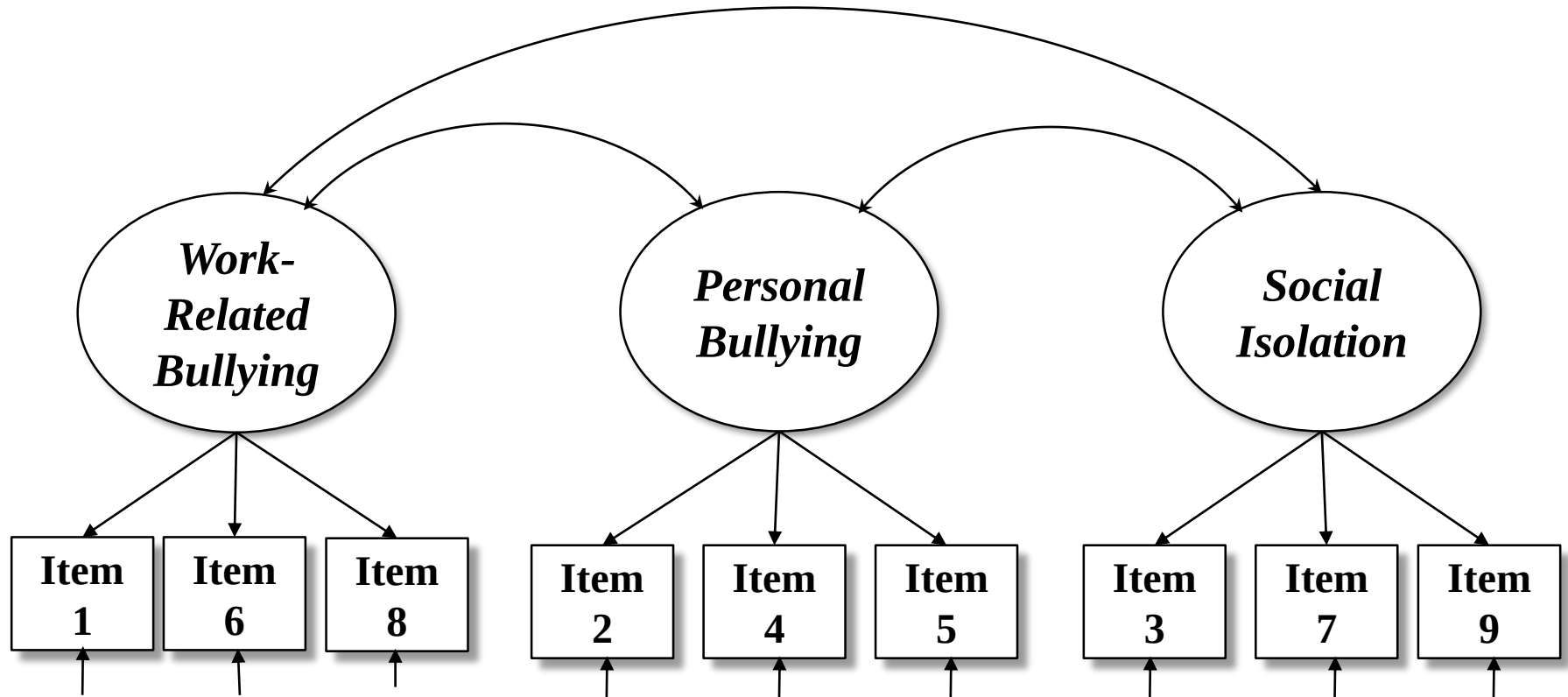
Use MI with caution: They must be **justified** from a theoretical and methodological point of view



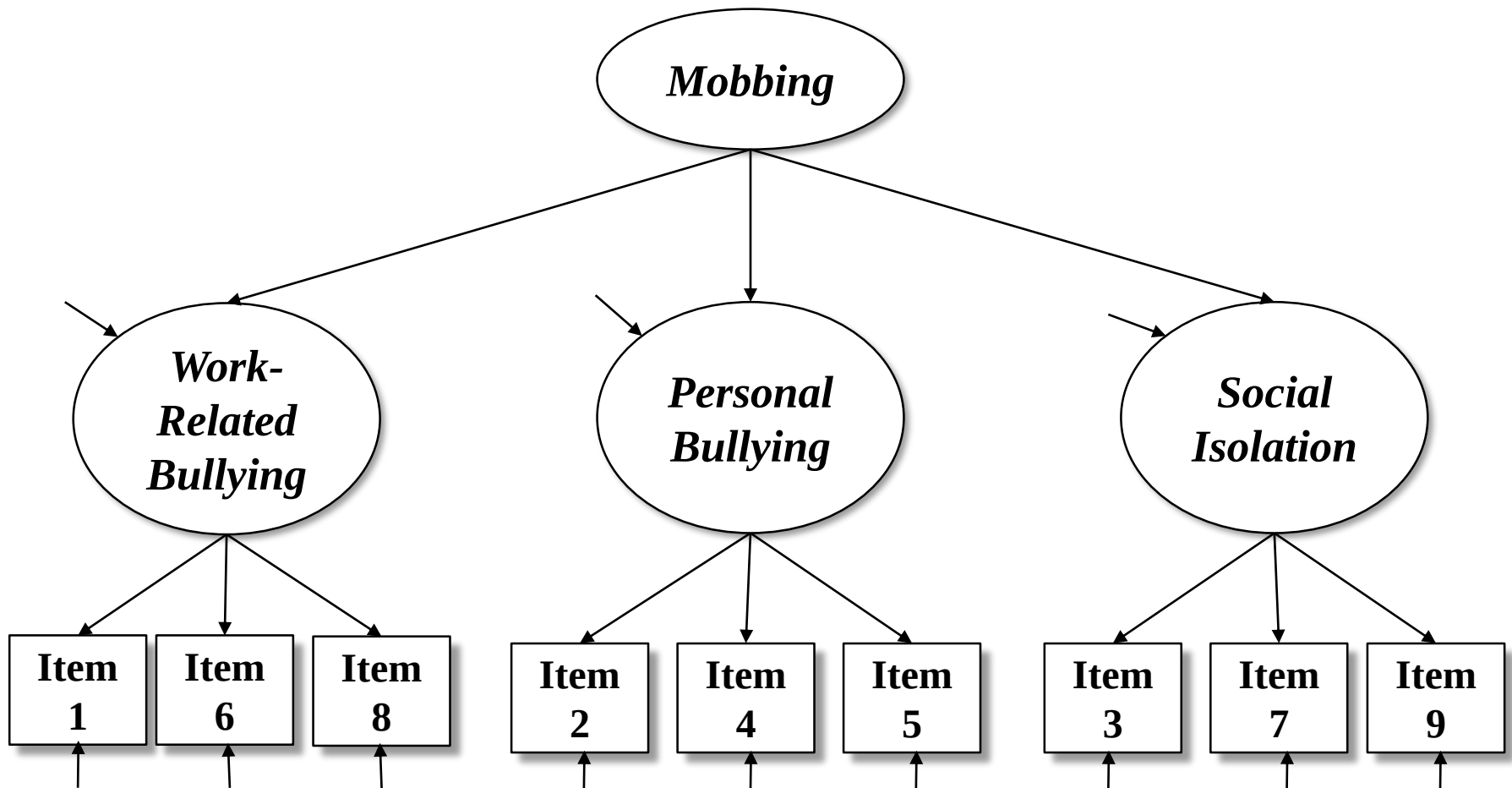
Structural Equation Modeling

Some Types and Applications

Confirmatory Factor Analysis (CFA)

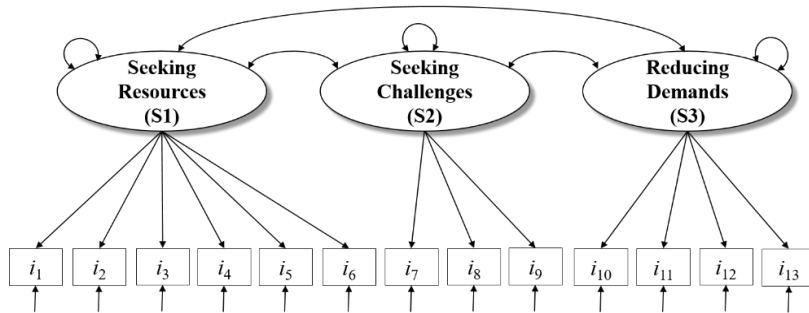


Higher-order (or Second-order) CFA

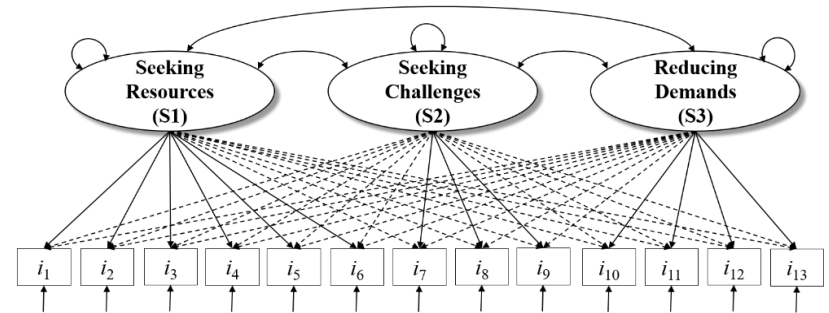


Multidimensional Structure

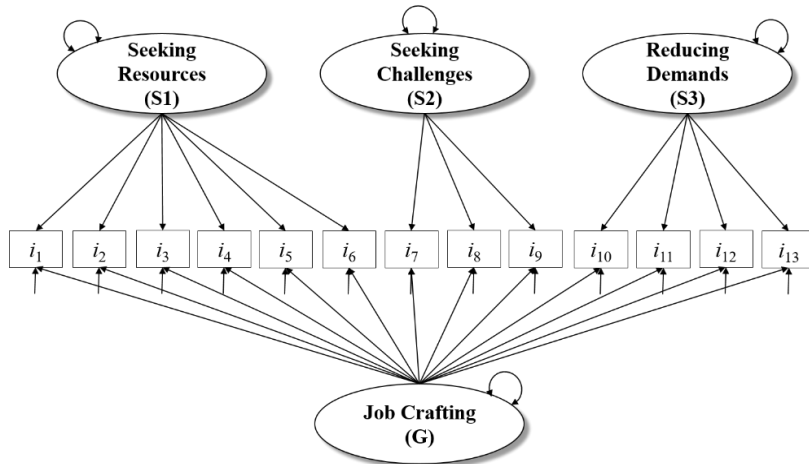
ICM-CFA



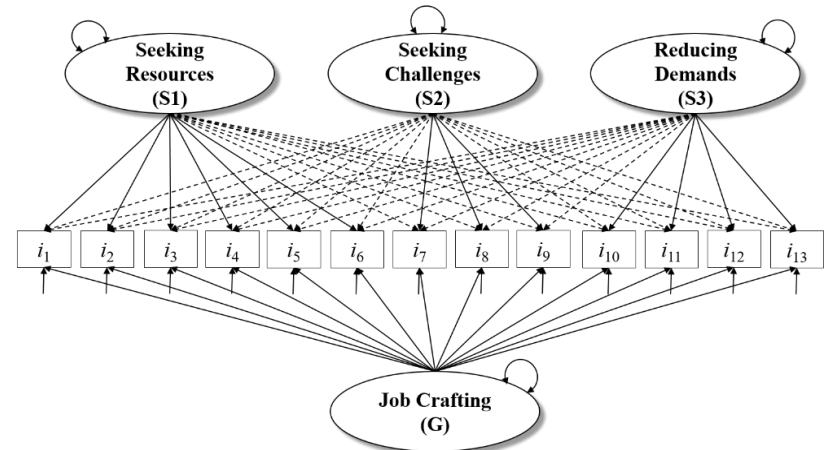
ESEM



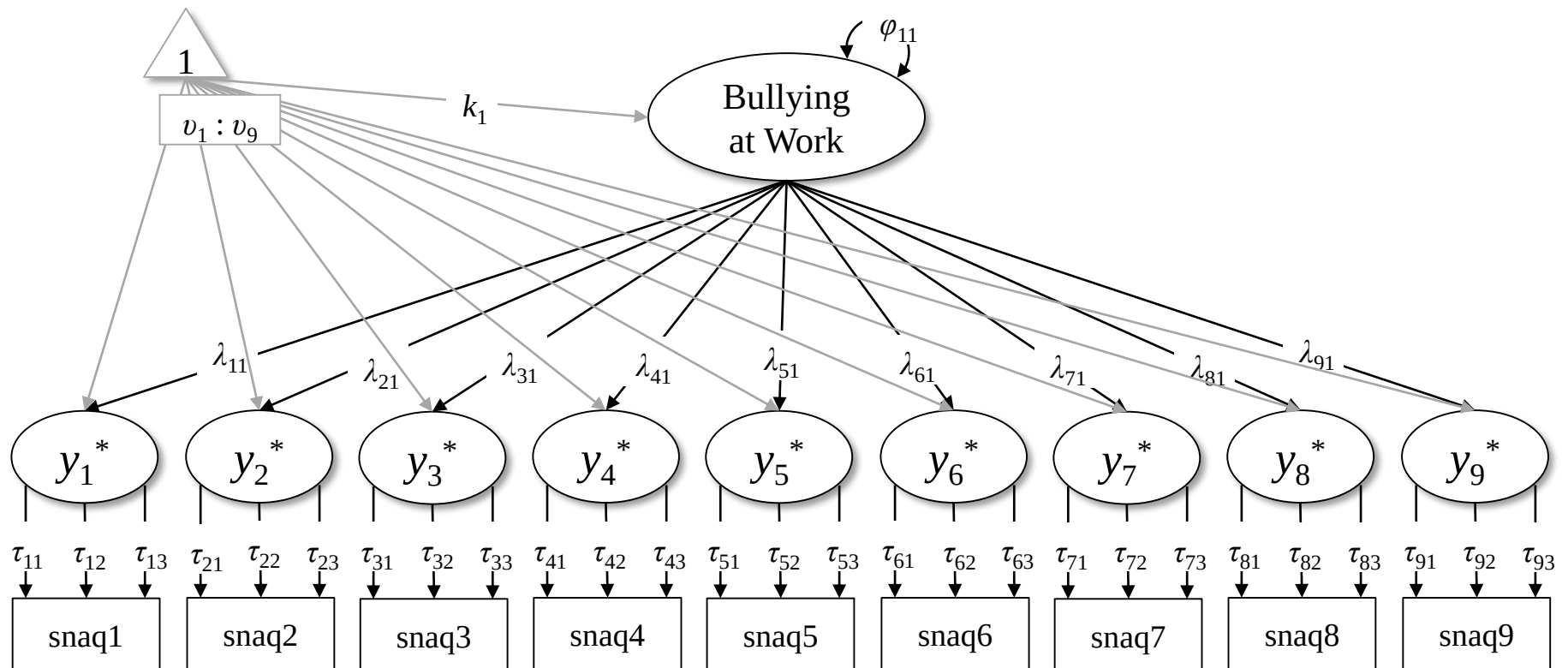
B-CFA



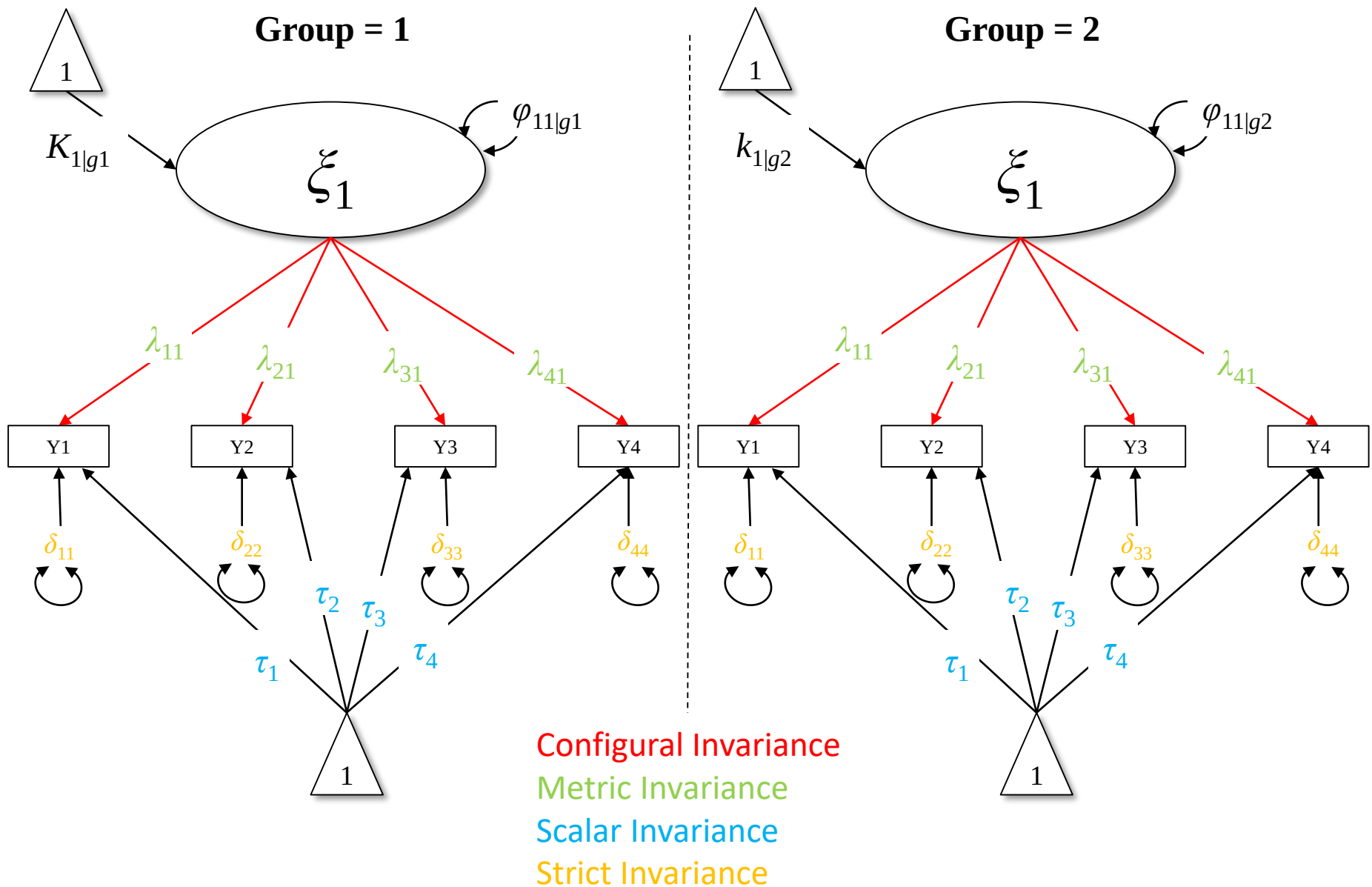
B-ESEM



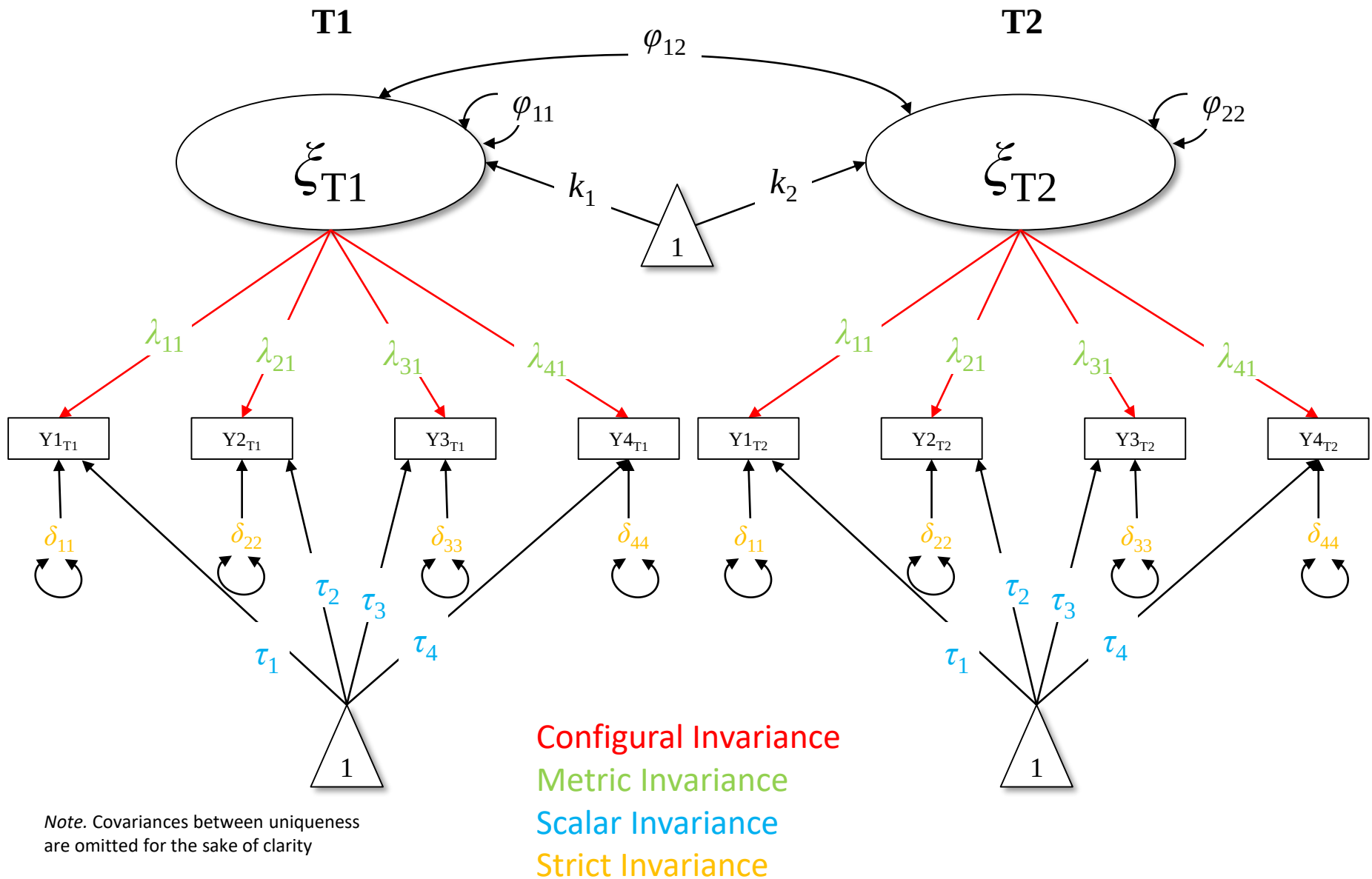
CFA with Ordinal Data



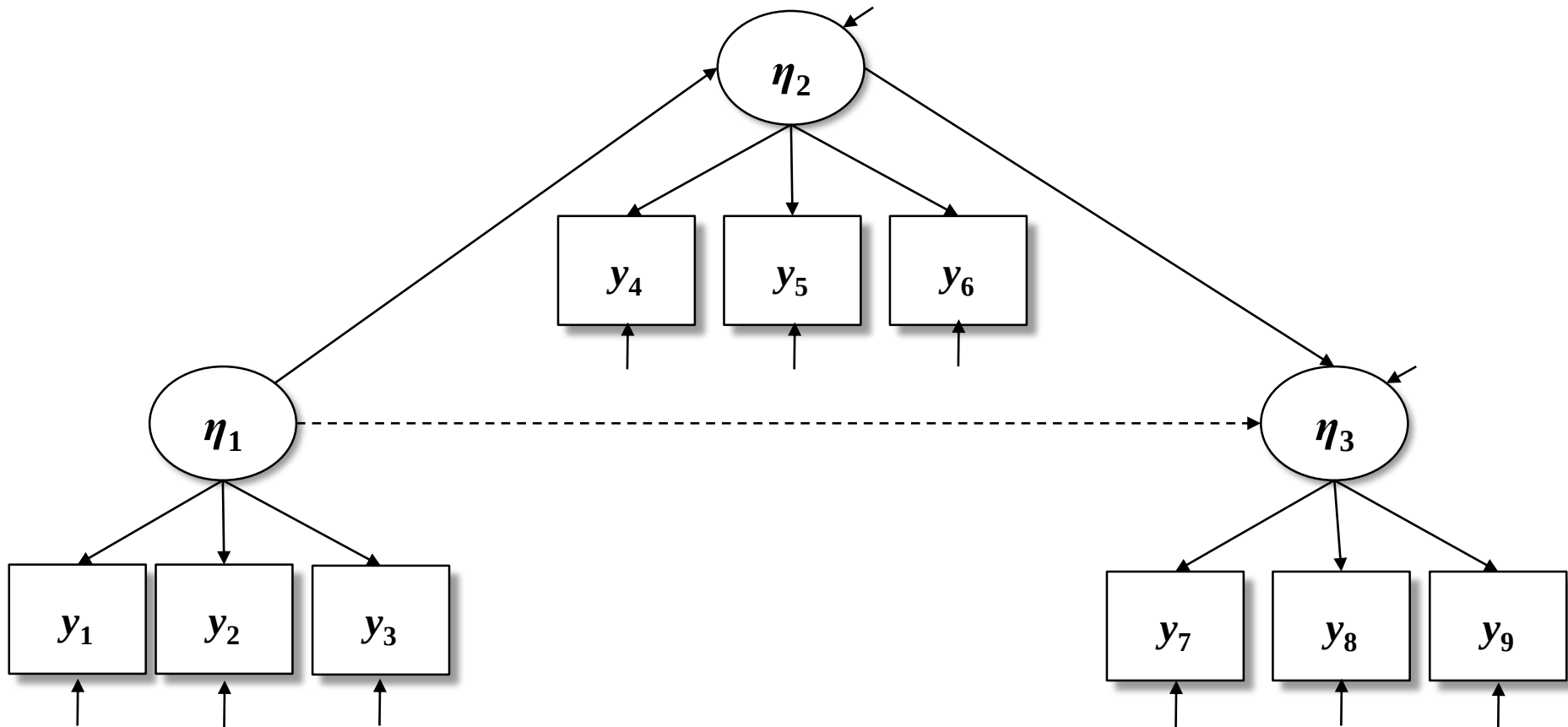
Measurement Invariance (Multiple-group)



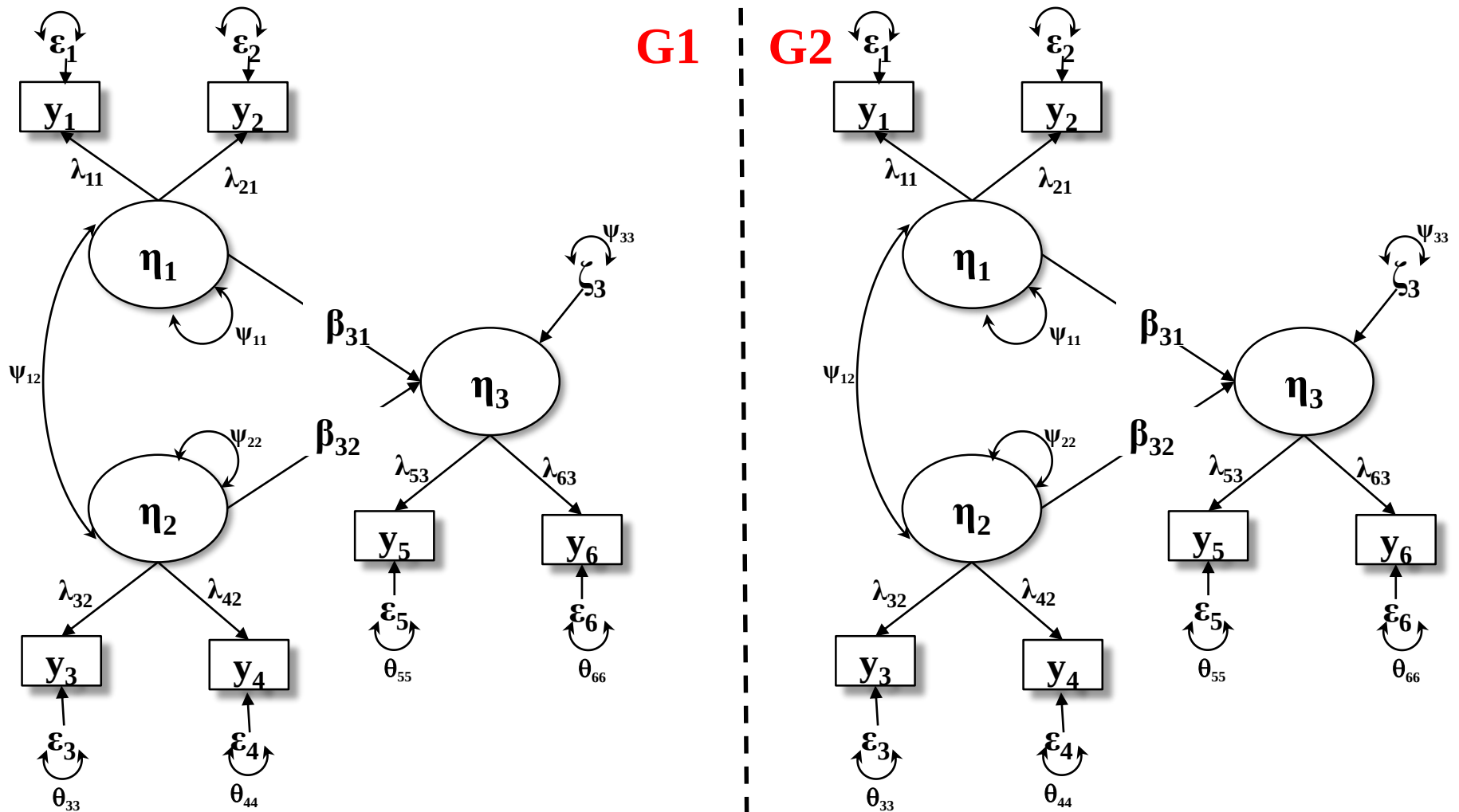
Measurement Invariance (Longitudinal)



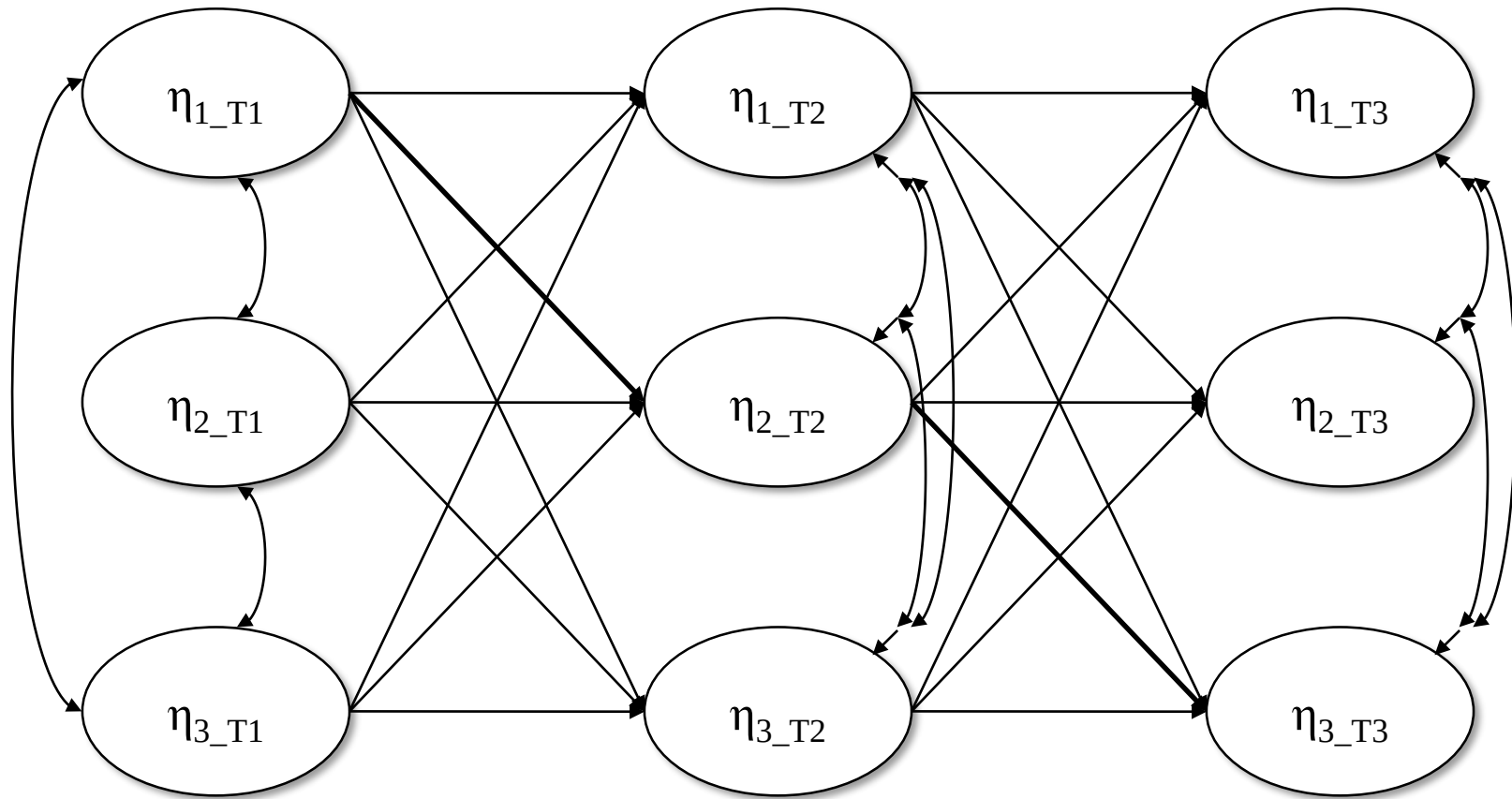
Cross-sectional Mediation with Latent Variables



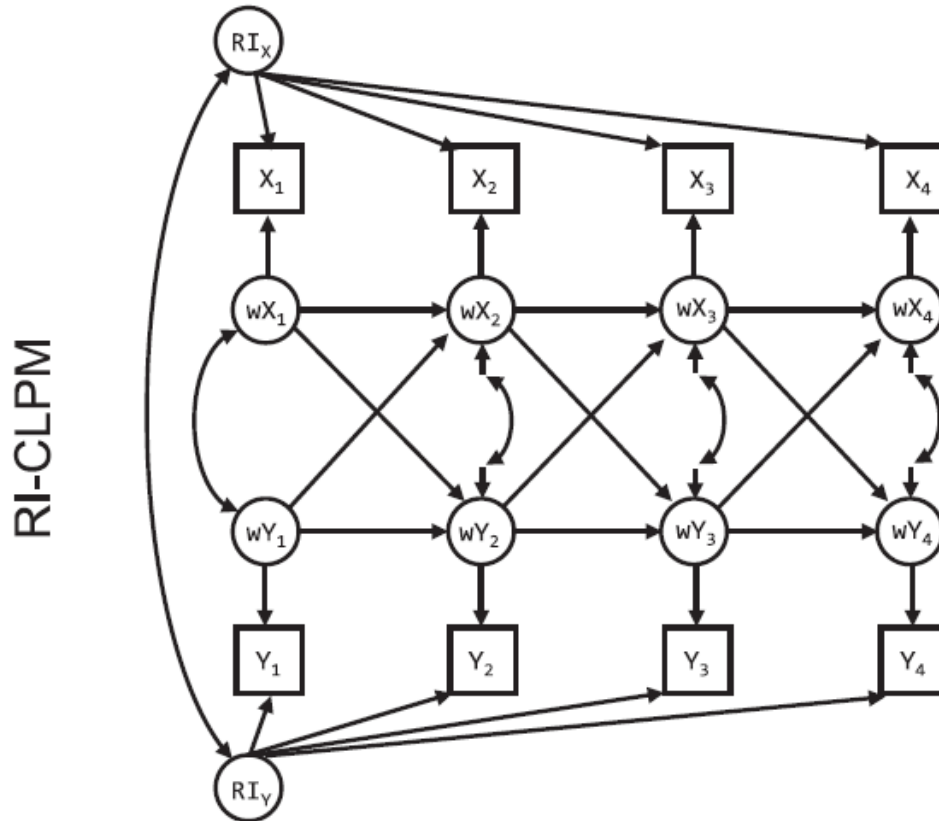
Multiple-group SEM (Jöreskog, *Psychometrika*, 1971)



Autoregressive Cross-Lagged Panel Model

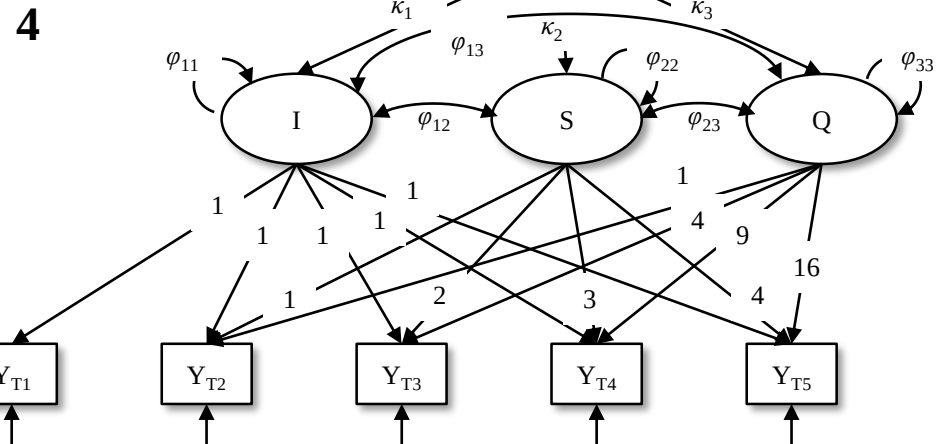
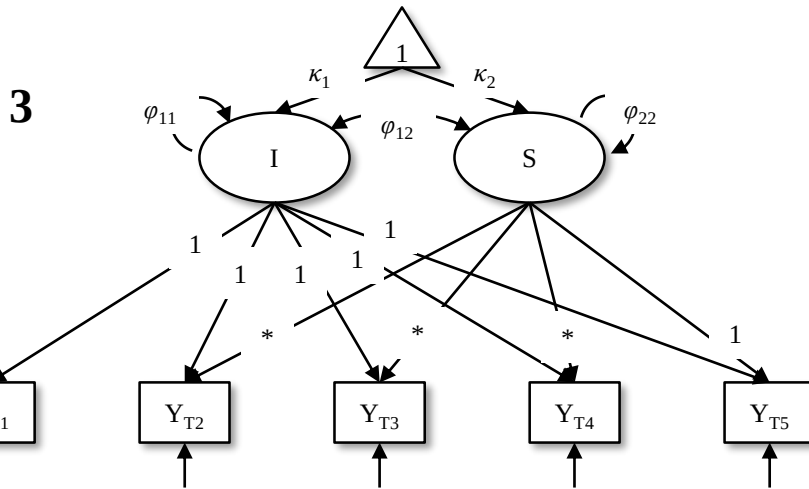
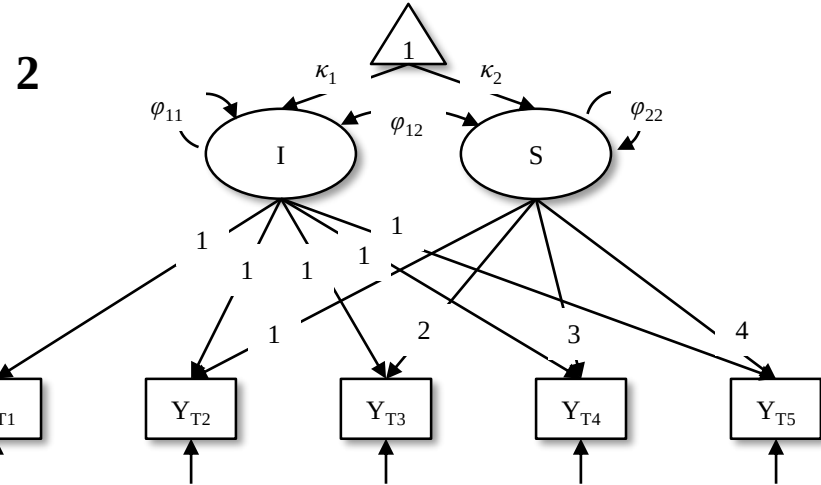
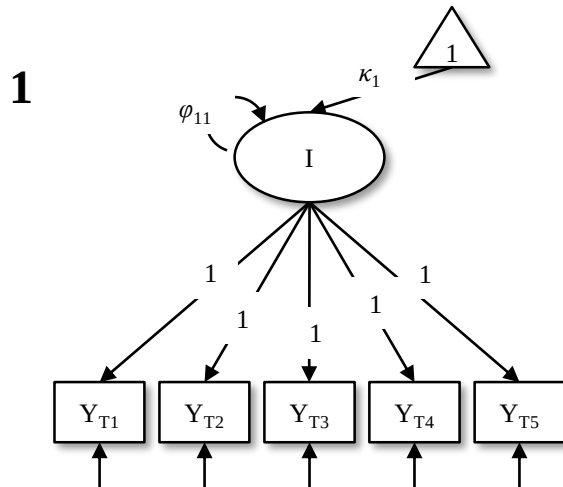


Random-Intercept CLPM (RI-CLPM)



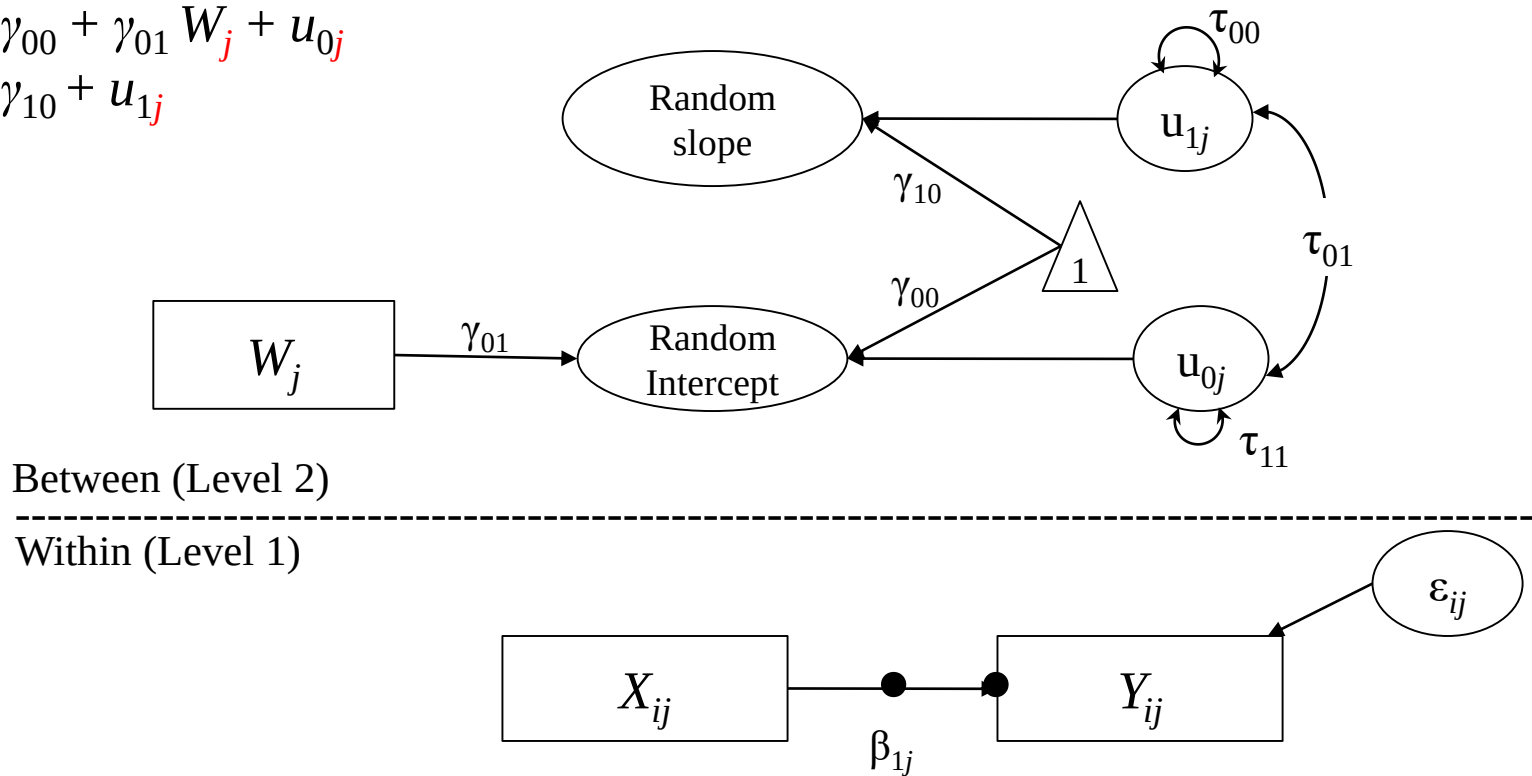
Note. Figure from Hamaker ([2026](#)). For more details on RI-CLPM and its dispute with classical ACLPM, see also Hamaker et al. ([2015](#)) and Lucas ([2023](#)). For a more theoretical contribution, see Hamaker ([2012](#)) and Borsboom & Haslbeck ([2024](#))

Latent Growth Models (Meredith & Tisak, *Psychometrika*, 1990)



Multilevel SEM

Level 1: $Y_{ij} = \beta_{0j} + \beta_{1j} X_{ij} + \varepsilon_{ij}$
 Level 2: $\beta_{0j} = \gamma_{00} + \gamma_{01} W_j + u_{0j}$
 Level 2: $\beta_{1j} = \gamma_{10} + u_{1j}$



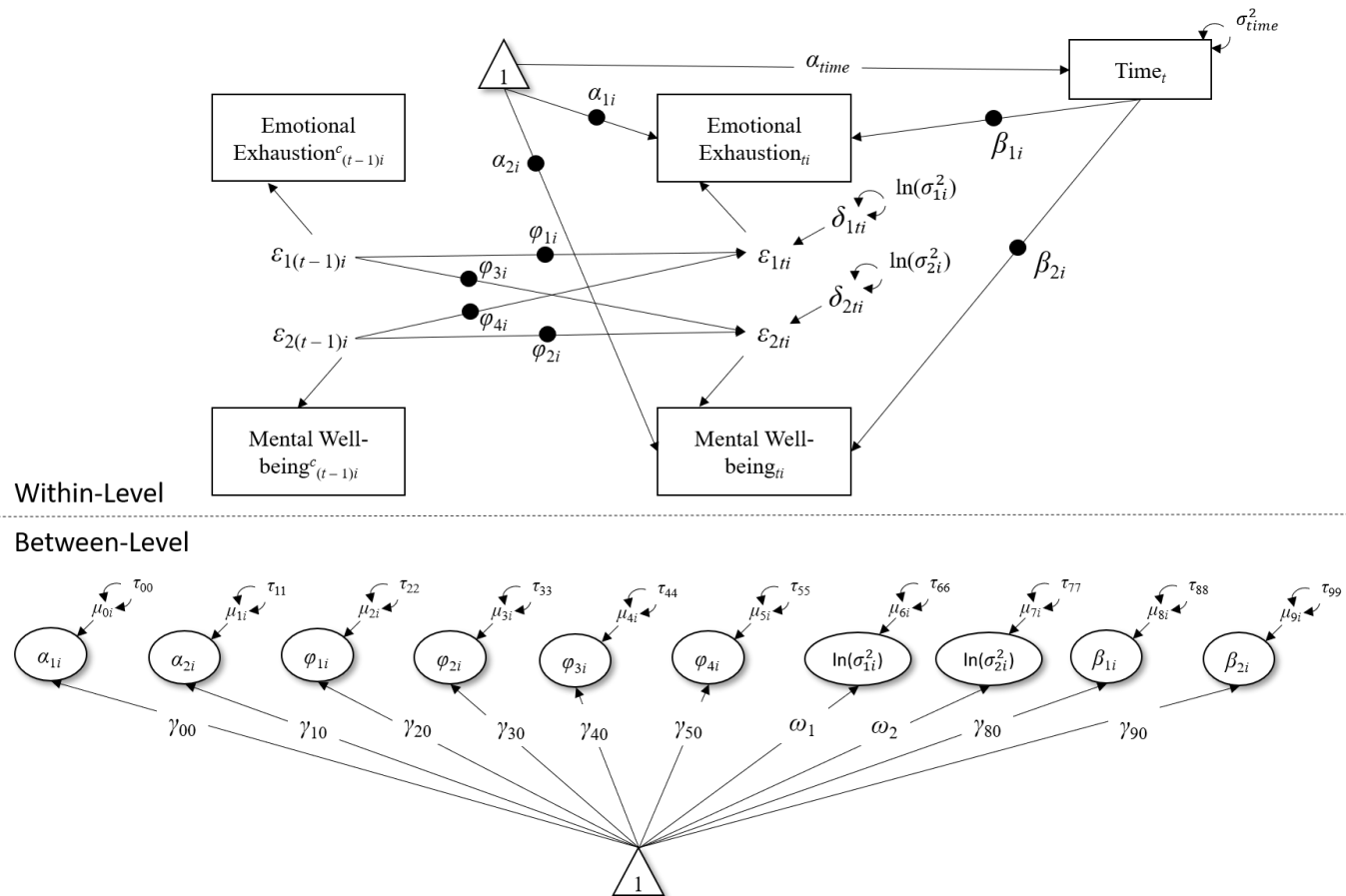
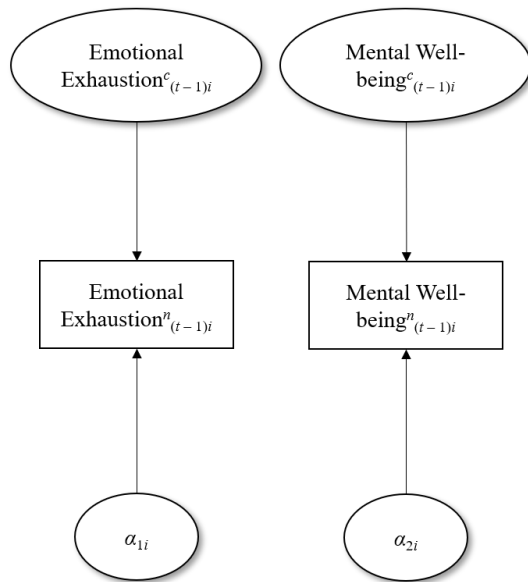
For sake of clarity, Formula and Figure refer to a simple multilevel regression.

The use of latent variables, or adding more than one dependent variable, would make this model a Multilevel SEM.

Dynamic SEM

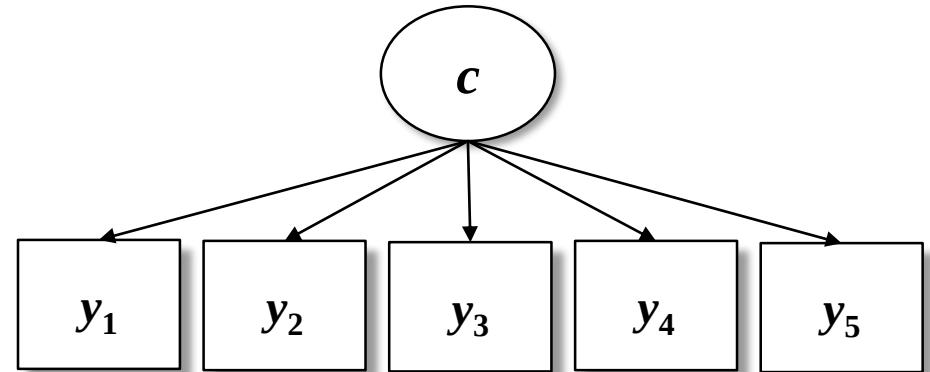
(Combination of Time Series Modeling, Multilevel Modeling, and SEM)

Latent Variable
Decomposition

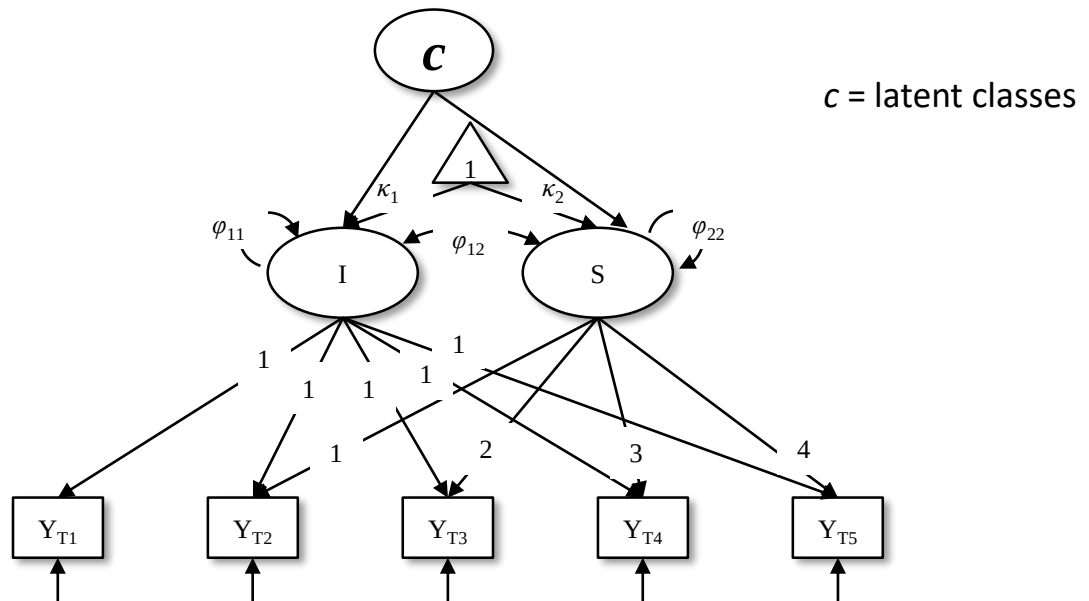


Mixture Models (Muthén, *Psychometrika*, 1989)

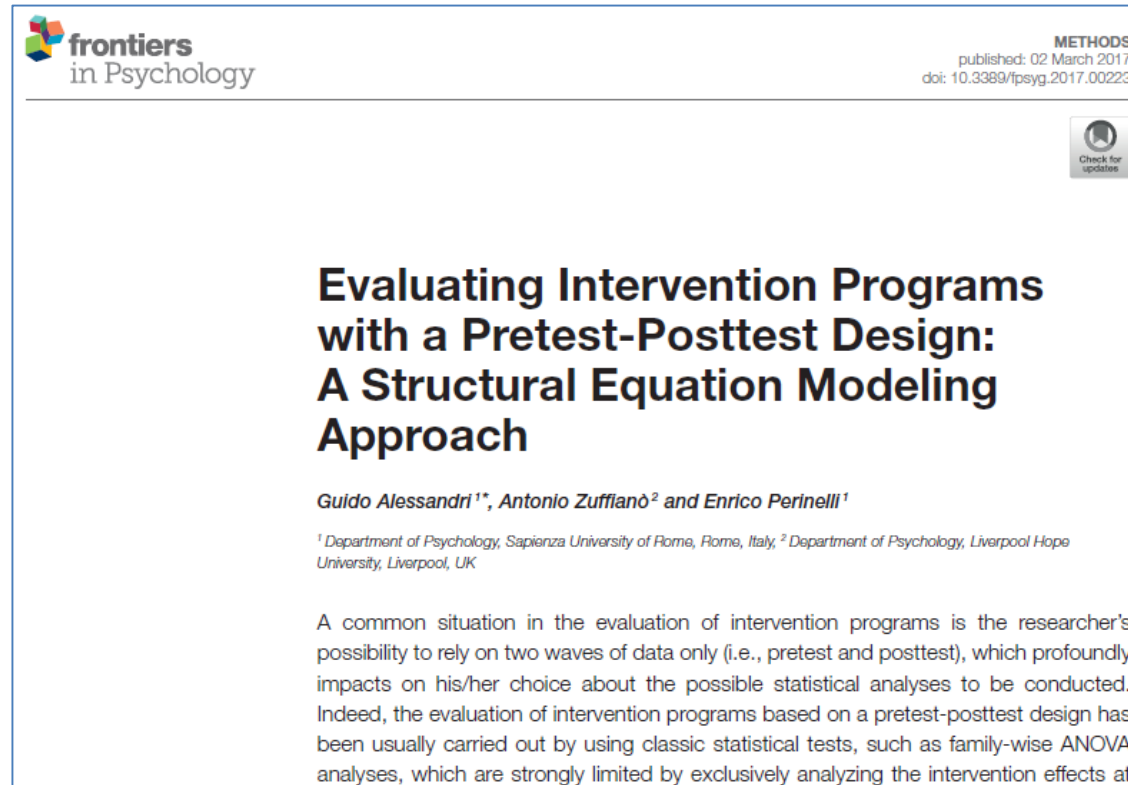
Latent Class Analysis (LCA; if y is dichotomous)
Latent Profile Analysis (LPA; if y is continuous)



Growth Mixture Model (GMM)



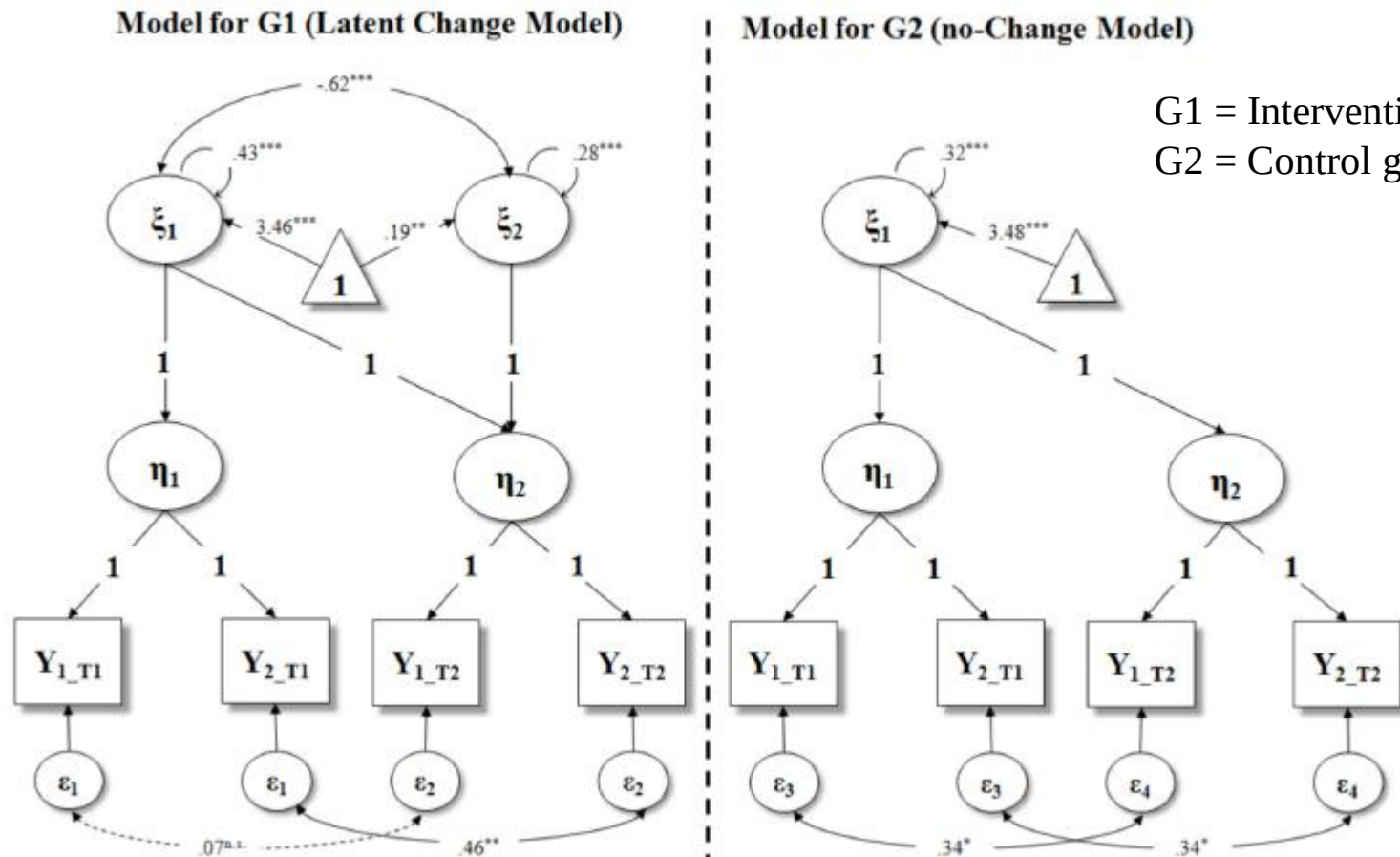
Advancing General Linear Models



Alessandri, G., Zuffianò, A., & Perinelli, E. (2017). Evaluating intervention programs with a pretest-posttest design: A Structural Equation Modeling approach. *Frontiers in Psychology: Quantitative Psychology and Measurement*, 8, Article 223.
<https://doi.org/10.3389/fpsyg.2017.00223>

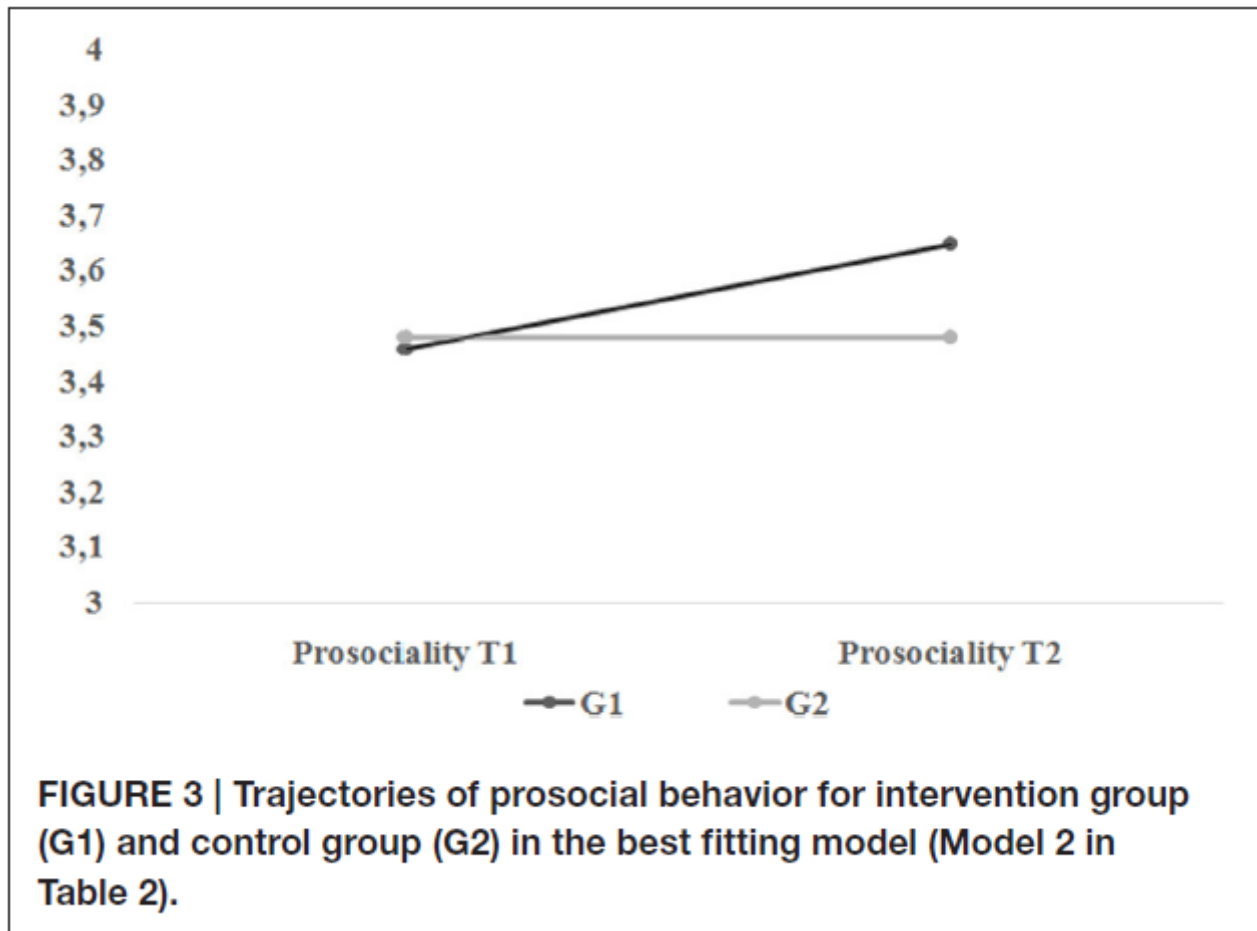
Advancing General Linear Models

Second-Order Multiple-Group Latent Curve Model



Advancing Mixed Models

Second-Order Multiple-Group Latent Curve Model



G1 = Intervention group
G2 = Control group



Other Types

- 1) Latent Transition Analysis
- 2) Latent Change (Difference) Score
- 3) Latent State-Trait (LST) Models

-
-
-
-
-

PhD Thesis	
On the Validity of the State-Like Component of Global Self-Esteem: Relationships with Implicit Self-Esteem and Work-Related Variables Using Different Latent State-Trait Models	
Supervisor	PhD Candidate
Professor Guido Alessandri	Enrico Perinelli
February 2018	

and much else....



Introduction to *Mplus* and *lavaan*



Mplus

Mplus is currently one of the most used software for latent variables analyses.

Mplus

Copyright © 1998-2017 Muthén & Muthén
Program Copyright © 1998-2017 Muthén & Muthén
Version 8
April 2017

Muthén & Muthén
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Los Angeles, CA 90066
Tel: (310) 391-9971
Fax: (310) 391-8971
Web: www.StatModel.com
Support@StatModel.com



Mplus

Its popularity is mainly due to:
a user-friendly syntax

<i>Syntax</i>	<i>Action</i>
ON	Regressed on (defines structural paths)
BY	Measured by (defines factor loadings)
WITH	Covaried with
IND	Indirect effect
[]	Mean structure and intercept
()	Assigns a label
@	Fixes a parameter to a specific value
*	Parameter freed/Specifies a starting value for a parameter
MODEL INDIRECT	Specifies the indirect effect to be computed
MODEL CONSTRAINT	Specifies linear and nonlinear constraints



Mplus

Its popularity is mainly due to:

The availability of a web-site (<https://www.statmodel.com/>) that provides:

1. information on the use of the software through a full-detailed user's guide (https://www.statmodel.com/download/usersguide/MplusUserGuideVer_8.pdf)
2. A forum in which users can deal (by questioning, by answering or simply by reading) with a broad variety of topics (<https://www.statmodel.com/cgi-bin/discus/discus.cgi>)
3. A plethora of real data examples (<https://www.statmodel.com/ugexcerpts.shtml>)
4. Videos and Handouts for Short Courses (https://www.statmodel.com/course_materials.shtml)



Mplus

Its popularity is mainly due to:

The availability of several estimators.

<i>Syntax</i>	<i>Estimator</i>
ML	Maximum Likelihood
MLM	Satorra-Bentler chi-square
MLMV	ML with mean- and variance-adjusted
MLR	Yuan-Bentler chi-square
MLF	ML with standard errors approximated by first-order derivatives
MUML	Muthén's limited information parameter estimates
WLS	Weighted least square
WLSM	Weighted least square mean-adjusted
WLSMV	Weighted least square mean- and variance-adjusted
ULS	Unweighted least squares
ULSMV	ULS mean- and variance-adjusted
GLS	Generalized Least Square
BAYES	Bayesian posterior parameter estimates with credibility intervals and posterior predictive checking



Mplus

Its popularity is mainly due to:

Constantly updated software

*Current version (June 2026): **9.1***

Version 9	October 2025
Version 8	April 2017
Version 7	September 2012
Version 6	April 2010
Version 5	November 2007
Version 4	February 2006
Version 3	March 2004
Version 2	February 2001
Version 1	November 1998

Mplus

Its popularity is mainly due to:
The availability of several books

- Byrne, B. M. (2013). *Structural equation modeling with Mplus: Basic concepts, applications, and programming*. Routledge.
- Geiser, C. (2013). *Data analysis with Mplus*. Guilford Press.
- Grimm, K. J. (2026). *Categorical Data Analysis with Structural Equation Models Applications in Mplus and lavaan*. Guilford Press.
- Grimm, K. J., Ram, N., & Estabrook, R. (2017). *Growth Modeling: Structural equation and multilevel modeling approaches*. Guilford Press.
- Heck, R. H., & Thomas, S. L. (2015). *An introduction to multilevel modeling techniques: MLM and SEM approaches using Mplus*. Routledge.
- Muthén, B., Muthén, L. K., Asparouhov, T. (2016). *Regression and mediation analysis using Mplus*. Los Angeles, CA: Muthén & Muthén.
- Peugh, J., & Litson, K. (2026). *Monte Carlo Simulation Power Analysis Using Mplus and R*. Guilford Press.
- Wang, J., & Wang, X. (2020). *Structural equation modeling: Applications using Mplus* (2nd ed.). John Wiley & Sons.
- Wickrama, K. A. S., Lee, T. K., O'Neal, C. W., & Lorenz, F. O. (2022). *Higher order growth curves and mixture modeling with Mplus: A practical guide* (2nd ed.). Routledge.

Mplus

Important steps for preparing dataset to be used in *Mplus*:

- 1) Coding missing values (I suggest -99 if you work in SPSS; for R, see the subsequent slide)
- 2) Create an ASCII data file and eventually convert commas with dots (in Notepad; for R this is not necessary)
- 3) Commands and Basic analysis (in *Mplus*)
- 4) Output for Basic analysis (in *Mplus*)



Mplus

Important steps for preparing dataset to be used in Mplus:

In order to create a dataset for Mplus, for **R** use the subsequent syntax

```
# Initial setup -----  
  
rm(list = ls())  
setwd("C:/Users/Enrico Perinelli/Desktop") # set your working directory or use a .Rproj file  
  
# Load or extract dataset -----  
  
library(psych)  
my_bfi <- bfi  
  
# Load `MplusAutomation` package -----  
  
library(MplusAutomation)  
  
# Use the `prepareMplusData` function -----  
  
prepareMplusData(  
  my_bfi,  
  filename = "bfi.dat"  
)
```

Mplus

big_five - Blocco note

File	Modifica	Formato	Visualizza	?	
404	3,50	3,63	4,38	4,50	1,13
311	4,00	4,75	4,75	4,13	2,38
379	2,88	3,50	3,75	3,88	1,38
338	4,38	3,75	4,00	3,00	1,13
209	3,75	4,38	4,13	4,25	1,63
141	-99,00	5,00	5,00	5,00	4,50
5	4,00	4,00	4,00	3,88	1,50
83	3,13	3,13	4,25	3,75	1,25
106	3,88	2,63	3,88	4,00	1,38
409	3,75	4,13	-99,00	-99,00	1,00
414	3,63	4,00	4,13	4,00	1,25
312	3,38	3,13	4,00	3,13	1,75
54	4,00	3,38	4,88	4,75	-99,00
296	3,00	3,00	3,00	3,00	3,00
411	4,38	4,00	4,88	4,50	1,00
343	4,25	4,13	5,00	4,00	1,00
287	2,88	3,00	3,00	3,00	2,63
369	4,00	-99,00	4,63	4,63	1,25
65	-99,00	4,63	4,50	4,50	1,00
89	3,50	4,63	4,13	3,38	1,25
347	3,13	3,50	-99,00	4,25	2,50
96	3,63	2,00	4,25	4,13	1,00
206	4,75	3,50	5,00	4,88	-99,00
142	1,00	1,00	1,00	1,00	1,50
144	4,00	5,00	5,00	5,00	3,50
143	4,00	4,00	4,00	4,00	3,75
324	3,63	3,38	4,50	4,00	1,13

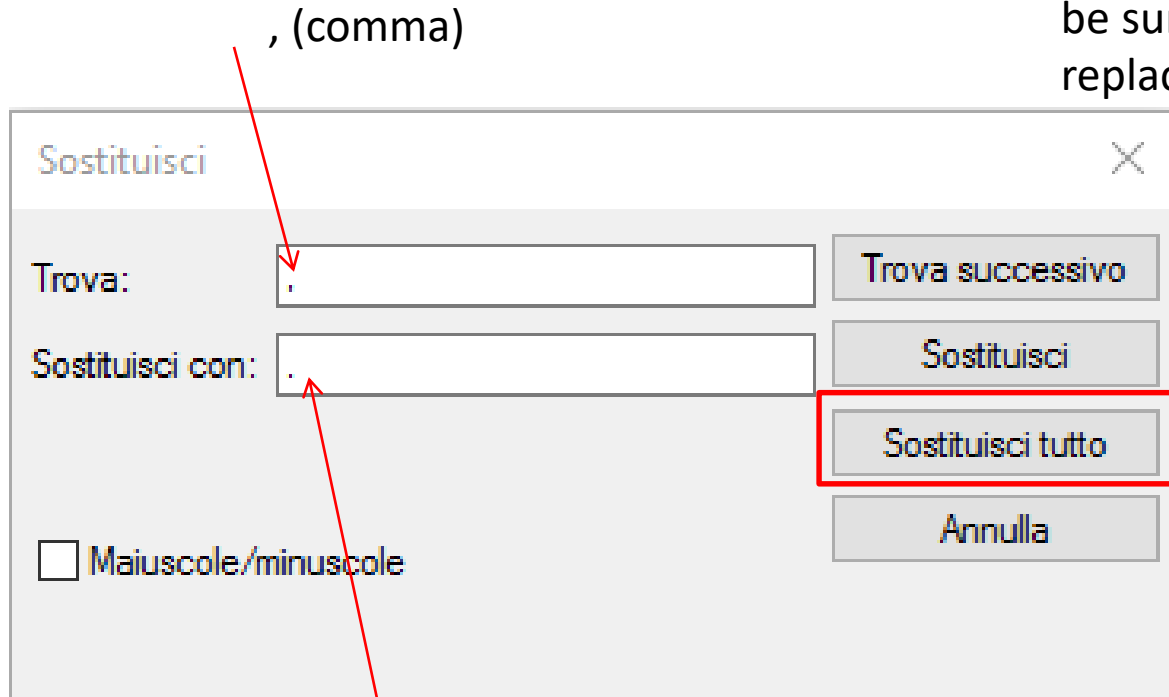
For **SPSS**, instead, you must be sure that commas are replaced with dots (1)

Modifica	Formato	Visualizza	?
Annulla		CTRL+Z	
Taglia		CTRL+X	
Copia		CTRL+C	
Incolla		CTRL+V	
Elimina		CANC	
Trova...		CTRL+T	
Trova successivo		F3	
Sostituisci...		CTRL+H	
Vai a...		CTRL+G	
Seleziona tutto		CTRL+A	
Ora/Data		F5	

Mplus

For **SPSS**, instead, you must
be sure that commas are
replaced with dots (2)

, (comma)



Sostituisci

Trova: ,

Sostituisci con: .

☐ Maiuscole/minuscole

Trova successivo

Sostituisci

Sostituisci tutto

Annulla

. (dot)

Mplus

For **SPSS**, instead, you must be sure that commas are replaced with dots (3)

big_five - Blocco note

File	Modifica	Formato	Visualizza	?		
404	3.50	3.63	4.38	4.50	1.13	
311	4.00	4.75	4.75	4.13	2.38	
379	2.88	3.50	3.75	3.88	1.38	
338	4.38	3.75	4.00	3.00	1.13	
209	3.75	4.38	4.13	4.25	1.63	
141	-99.00	5.00	5.00	5.00	4.50	
5	4.00	4.00	4.00	3.88	1.50	
83	3.13	3.13	4.25	3.75	1.25	
106	3.88	2.63	3.88	4.00	1.38	
409	3.75	4.13	-99.00	-99.00	1.00	
414	3.63	4.00	4.13	4.00	1.25	
312	3.38	3.13	4.00	3.13	1.75	
54	4.00	3.38	4.88	4.75	-99.00	
296	3.00	3.00	3.00	3.00	3.00	
411	4.38	4.00	4.88	4.50	1.00	
343	4.25	4.13	5.00	4.00	1.00	
287	2.88	3.00	3.00	3.00	2.63	
369	4.00	-99.00	4.63	4.63	1.25	
65	-99.00	4.63	4.50	4.50	1.00	
89	3.50	4.63	4.13	3.38	1.25	
347	3.13	3.50	-99.00	4.25	2.50	
96	3.63	2.00	4.25	4.13	1.00	
206	4.75	3.50	5.00	4.88	-99.00	
142	1.00	1.00	1.00	1.00	1.50	
144	4.00	5.00	5.00	5.00	3.50	
143	4.00	4.00	4.00	4.00	3.75	

NOW, ALL THE
COMMAS ARE
DOTS



Mplus

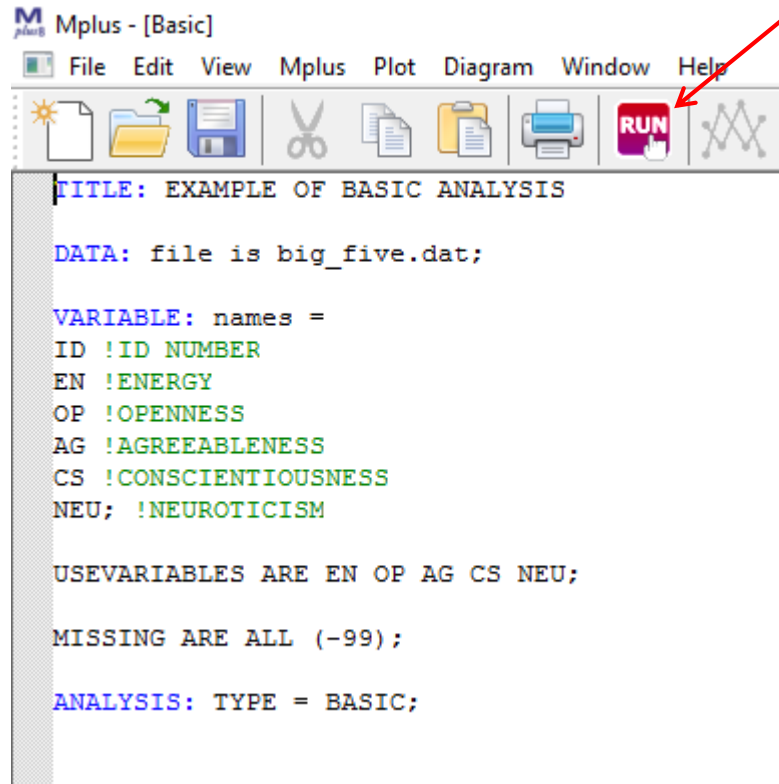
<i>Syntax</i>	<i>Function</i>
TITLE	Provide a title for the analysis
DATA	Provide information about the data set to be analyzed
VARIABLE	Provide information about the variables in the data set
DEFINE	Transform existing variables and create new variables
ANALYSIS	Describe the technical details of the analysis (e.g., Estimator, type)
MODEL	Describe the model to be estimated
OUTPUT	Request additional output not included as the default
SAVEDATA	Save the analysis data (e.g., Auxiliary data)
PLOT	Request graphical displays of analysis results
MONTECARLO	Specify the details of a monte carlo simulation study

! This symbol is used to take notes

; This symbol must be placed at the end of each command line

Mplus

CLICK RUN

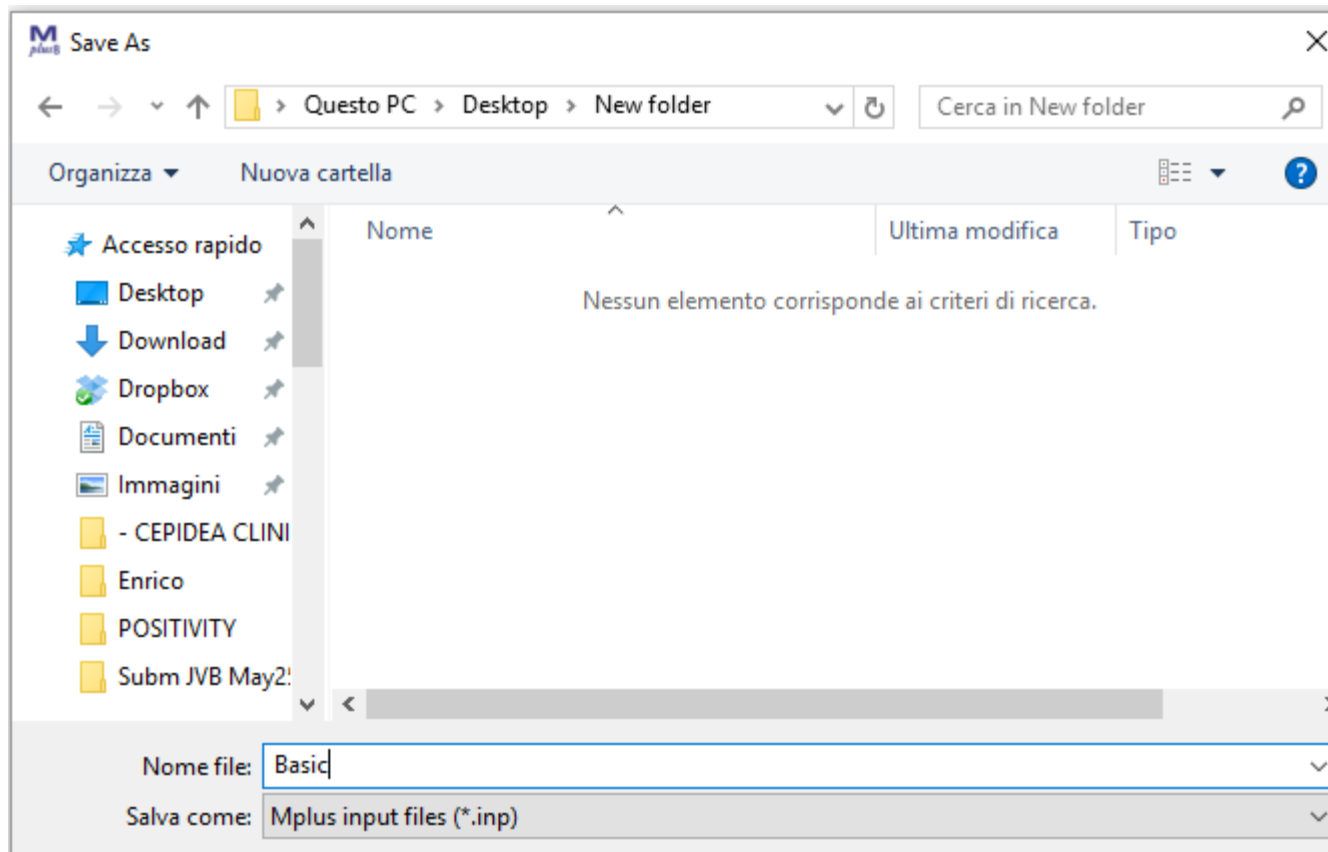


```
TITLE: EXAMPLE OF BASIC ANALYSIS  
  
DATA: file is big_five.dat;  
  
VARIABLE: names =  
ID !ID NUMBER  
EN !ENERGY  
OP !OPENNESS  
AG !AGREEABLENESS  
CS !CONSCIENTIOUSNESS  
NEU; !NEUROTICISM  
  
USEVARIABLES ARE EN OP AG CS NEU;  
  
MISSING ARE ALL (-99);  
  
ANALYSIS: TYPE = BASIC;
```

Remember that
variable names cannot
be longer than
8 characters

Mplus

Save the input file in the same folder of the .dat file



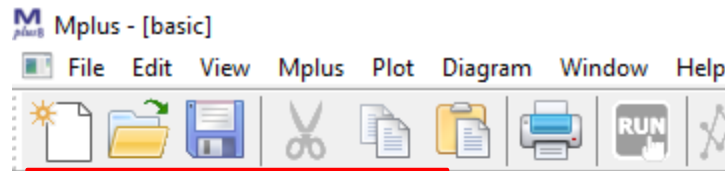


Mplus

New folder				
	Nome	Ultima modifica	Tipo	Dimensione
teacher	 Basic	21/08/2017 16:36	File INP	1 KB
	 basic	21/08/2017 16:36	File OUT	6 KB
erlinelli	 big_five	21/08/2017 16:37	File DAT	21 KB
C				
enti				

Mplus

Mplus version
and date



```
Mplus VERSION 8  
MUTHEN & MUTHEN  
08/21/2017 4:36 PM
```

Input instruction

```
INPUT INSTRUCTIONS  
  
TITLE: EXAMPLE OF BASIC ANALYSIS  
  
DATA: file is big_five.dat;  
  
VARIABLE: names =  
ID !ID NUMBER  
EN !ENERGY  
OP !OPENNESS  
AG !AGREEABLENESS  
CS !CONSCIENTIOUSNESS  
NEU; !NEUROTICISM  
  
USEVARIABLES ARE EN OP AG CS NEU;  
  
MISSING ARE ALL (-99);  
  
ANALYSIS: TYPE = BASIC;
```

Mplus

INPUT READING TERMINATED NORMALLY

The analysis has gone well!!

Title and information on the number of variables. Here we have only continuous variables, but this part of the output may display also the number of latent variables

EXAMPLE OF BASIC ANALYSIS

SUMMARY OF ANALYSIS

Number of groups	1
Number of observations	416
Number of dependent variables	5
Number of independent variables	0
Number of continuous latent variables	0
Observed dependent variables	
Continuous	
EN	OP AG CS NEU

Estimator	ML
Information matrix	OBSERVED
Maximum number of iterations	1000
Convergence criterion	0.500D-04
Maximum number of steepest descent iterations	20
Maximum number of iterations for H1	2000
Convergence criterion for H1	0.100D-03

By default in most of analyses the estimator is ML (in Multilevel modeling the default estimator is MLR)

Input data file(s)
big_five.dat

Input data format FREE

Mplus

COVARIANCE COVERAGE OF DATA

Minimum covariance coverage value 0.100

PROPORTION OF DATA PRESENT

	Covariance Coverage				
	EN	OP	AG	CS	NEU
EN	0.988				
OP	0.954	0.966			
AG	0.957	0.935	0.969		
CS	0.966	0.950	0.952	0.978	
NEU	0.962	0.945	0.945	0.952	0.974



This matrix shows the percentage of covariance coverage for each pair of observed variables. When dataset does not have missing data, each pair of observed variables has a covariance coverage of 100%.



Mplus

RESULTS FOR BASIC ANALYSIS

ESTIMATED SAMPLE STATISTICS

Means					
	EN	OP	AG	CS	NEU
	3.812	3.811	4.463	4.097	1.382
Covariances					
	EN	OP	AG	CS	NEU
EN	0.279				
OP	0.129	0.386			
AG	0.112	0.131	0.204		
CS	0.089	0.132	0.123	0.264	
NEU	-0.024	-0.004	-0.068	-0.012	0.264
Correlations					
	EN	OP	AG	CS	NEU
EN	1.000				
OP	0.392	1.000			
AG	0.470	0.469	1.000		
CS	0.330	0.412	0.531	1.000	
NEU	-0.087	-0.012	-0.294	-0.046	1.000

Here, information concerning Means, Covariances and Correlations are reported.



Mplus

UNIVARIATE SAMPLE STATISTICS

UNIVARIATE HIGHER-ORDER MOMENT DESCRIPTIVE STATISTICS

Variable/ Sample Size		Mean/ Variance	Skewness/ Kurtosis	Minimum/ Maximum	% with Min/Max	Percentiles		Median
						20%/60%	40%/80%	
EN	411.000	3.810	-0.390	1.000	0.24%	3.380	3.630	3.880
OP	402.000	3.809	-0.407	1.000	0.25%	3.250	3.750	3.880
AG	403.000	4.465	-1.496	1.000	0.25%	4.000	4.380	4.500
CS	407.000	4.098	-0.806	1.000	0.25%	3.630	4.000	4.130
NEU	405.000	1.382	2.439	1.000	29.63%	1.000	1.130	1.130
		0.264	8.499	4.500	0.49%	1.250	1.750	

Here, information concerning univariate sample statistics are reported.

A VERY IMPORTANT STEP IS TO CHECK THE TRUSTWORTHINESS OF THESE VALUES!!!!
WHEN THE DATASET IS LARGE, IT IS EASY TO WRONG SPECIFY VARIABLE NAMES, OR FORGET TO
SPECIFY THE MASSING DATA VALUE.



Mplus User's guide

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(types of SEM)

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commands and
syntaxes

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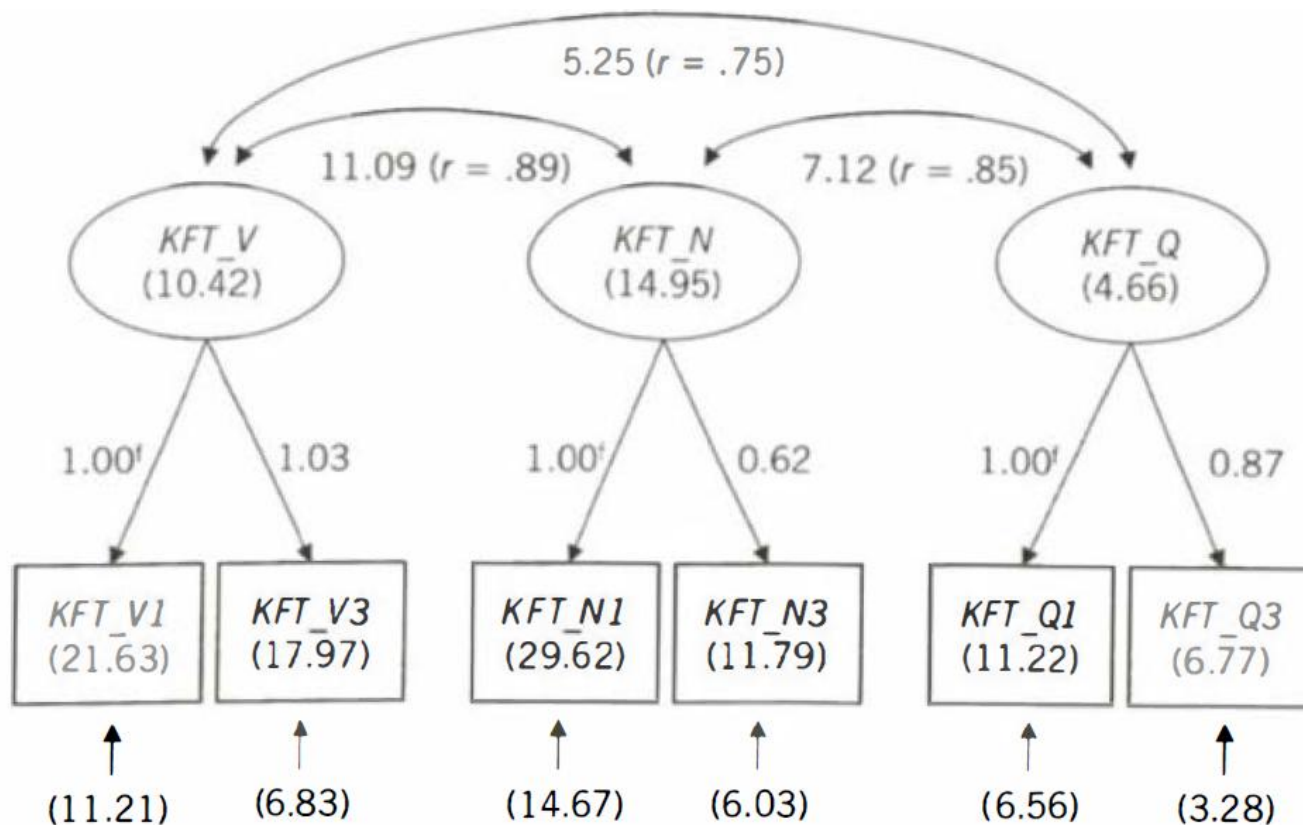
Links:

<https://www.statmodel.com/ugexcerpts.shtml>

https://www.statmodel.com/download/usersguide/MplusUserGuideVer_8.pdf

Mplus CFA example

(Geiser, 2013, p. 53)





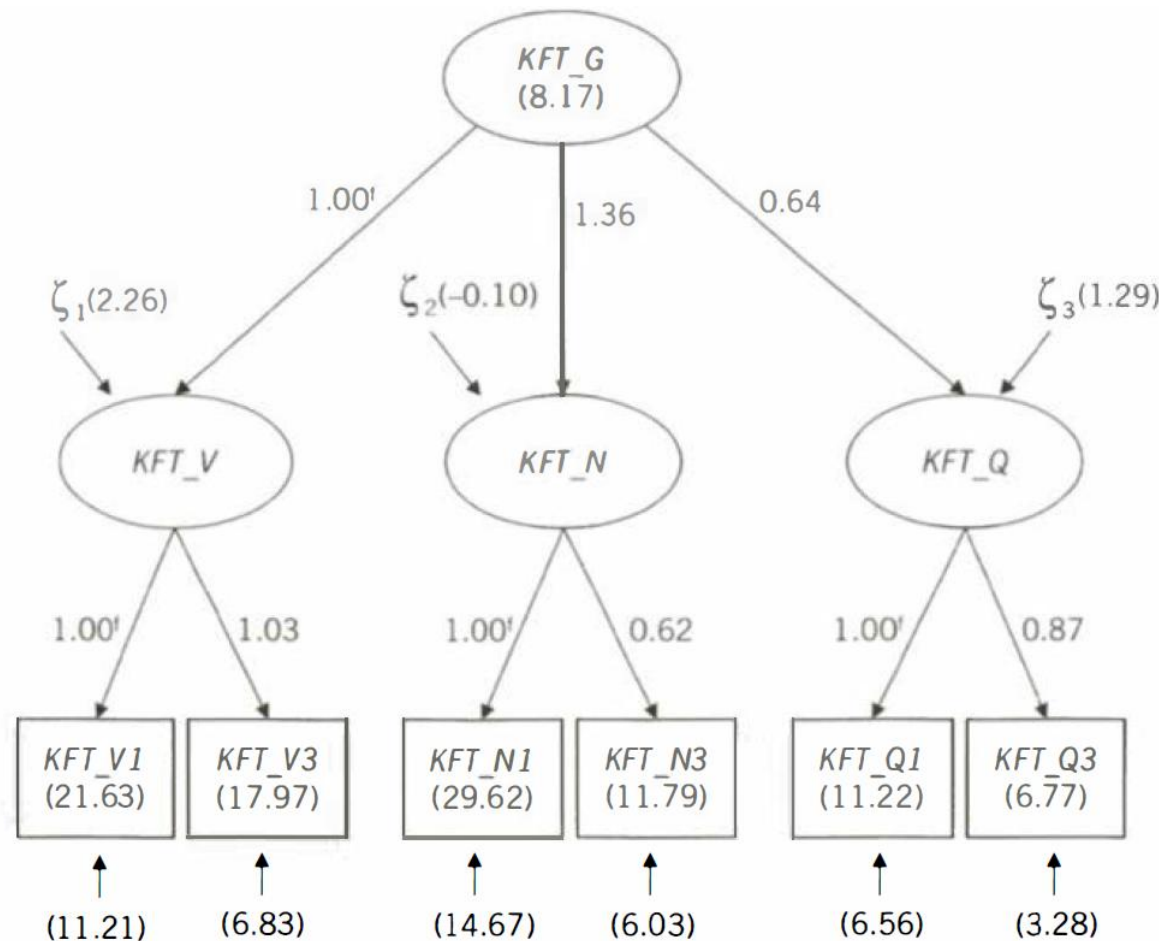
Mplus CFA example

(Geiser, 2013, p. 53)

```
title: CFA for the six KFT variables  
      3-Factor model (see Figure 3.8)  
  
data: file = KFT.dat;  
      listwise = on;  
  
variable: names = kft_v1 kft_v3 kft_q1 kft_q3 kft_n1 kft_n3;  
          missing = all(-99);  
  
analysis: type = general;  
  
model: KFT_V by kft_v1 kft_v3; ! verbal ability factor  
      KFT_Q by kft_q1 kft_q3; ! numeric ability factor  
      KFT_N by kft_n1 kft_n3; ! figural (non-verbal) ability factor  
  
output: sampstat stdyx tech4;
```

Mplus Second-Order CFA example

(Geiser, 2013, p. 59)



Mplus Second-Order CFA example

(Geiser, 2013, p. 59)

```
title: CFA for the six KFT variables
      Second-order factor model (see Figure 3.11)

data: file = KFT.dat;
      listwise = on;

variable: names = kft_v1 kft_v3 kft_q1 kft_q3 kft_n1 kft_n3;
          missing = all(-99);

analysis: type = general;

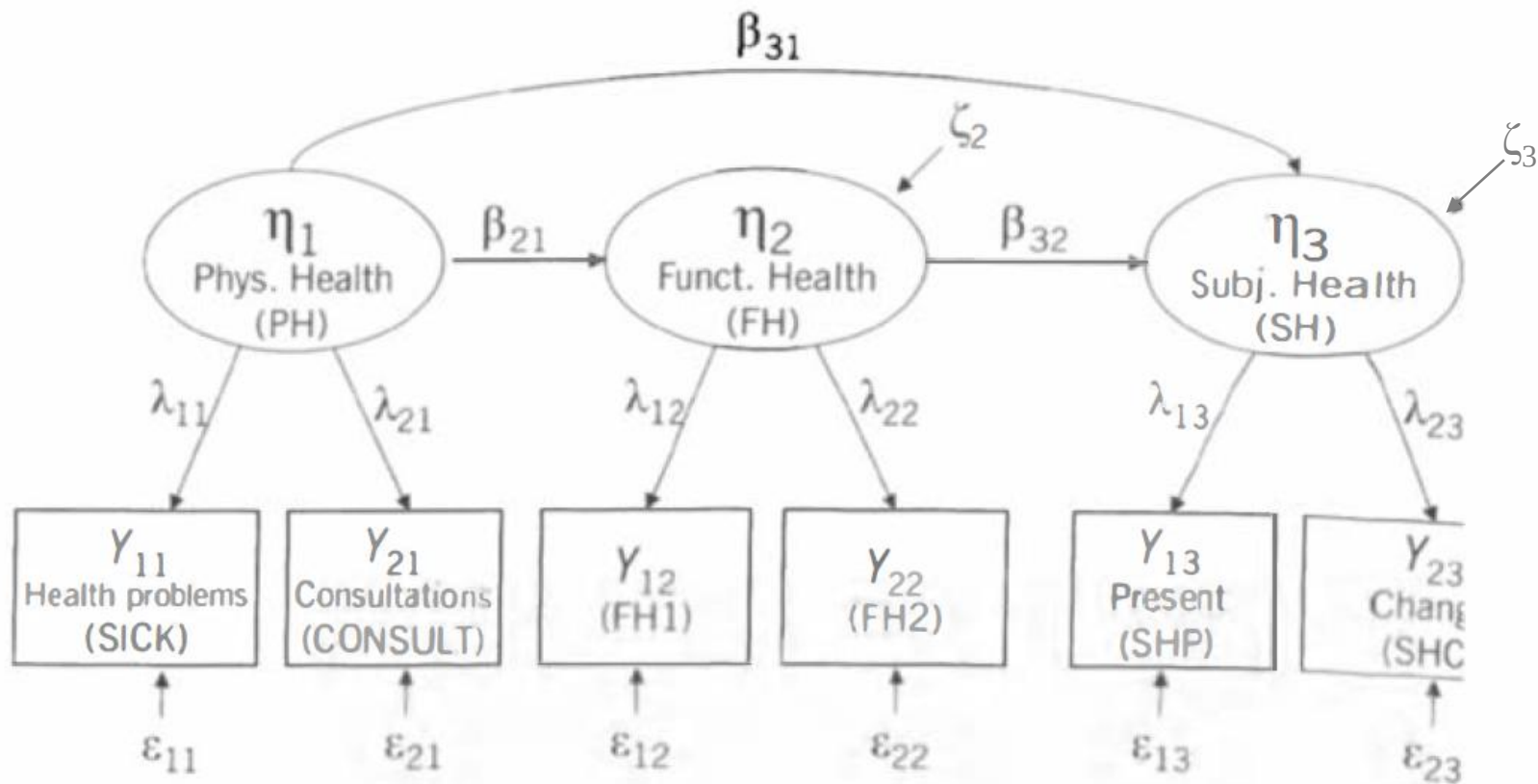
model: ! First-order factors
      KFT_V by kft_v1 kft_v3; ! verbal ability factor
      KFT_Q by kft_q1 kft_q3; ! numeric ability factor
      KFT_N by kft_n1 kft_n3; ! figural (non-verbal) ability factor

      ! Second-order factor
      KFT_G by KFT_V KFT_Q KFT_N;

output: sampstat stdyx;
```

Mplus SEM example

(Geiser, 2013, p. 74)





Mplus SEM example

(Geiser, 2013, p. 74)

```
title: Latent path model shown in Figure 3.15
      Relationship between physical health, functional health, and subjective health

data: file = health.dat;

variable: names = SHP ! Subjektive health in the present
              SHC ! Changes in subjective health across the past 6 years
              SICK ! Number of health issues in the past 12 months
              CONSULT ! Number of physician consultations in the past 12 months
              FH ! Total score SF-36 scale on functional health
              FH1 FH2; ! SF-36 scale split into 2 test halves/item parcels

      usevar = SHP SHC SICK CONSULT FH1 FH2;

analysis: type = general;

model: SH by SHP SHC; ! Factor/Measurement model subjective health
      PH by SICK CONSULT; ! Factor/Measurement model physical health
      FH by FH1 FH2; ! Factor/Measurement model functional health

! Path analysis at the level of the latent variables
      FH on PH;
      SH on PH FH;

      model indirect: SH ind PH; ! Request output of direct, indirect, and total effects

output: sampstat stdyx;
```



Mplus SEM example (bootstrap)

(Geiser, 2013, p. 74)

```
title: Latent path model shown in Figure 3.15
      Relationship between physical health, functional health, and subjective health
      With bias-corrected bootstrap confidence intervals

data: file = health.dat;

variable: names = SHP ! Subjektive health in the present
              SHC ! Changes in subjective health across the past 6 years
              SICK ! Number of health issues in the past 12 months
              CONSULT ! Number of physician consultations in the past 12 months
              FH ! Total score SF-36 scale on functional health
              FH1 FH2; ! SF-36 scale split into 2 test halves/item parcels

usevar = SHP SHC SICK CONSULT FH1 FH2;

analysis: type = general;
          bootstrap = 10000; ! Request 10000 bootstrap samples

model: SH by SHP SHC; ! Factor/Measurement model subjective health
       PH by SICK CONSULT; ! Factor/Measurement model physical health
       FH by FH1 FH2; ! Factor/Measurement model functional health

! Path analysis at the level of the latent variables
       FH on PH;
       SH on PH FH;

model indirect: SH ind PH; ! Request output of direct, indirect, and total effects

output: sampstat stdyx
        cinterval(bcbootstrap); ! Request confidence intervals based on bias-corrected bootstrap
```




lavaan

Material available at

https://github.com/EnricoPerinelli/Intro-to-SEM/blob/main/lavaan_example.R



UNIVERSITÀ
DI TRENTO

Dipartimento di
Psicologia e Scienze Cognitive

**Intro to
SEM**
Enrico Perinelli

Suggested Reading



Suggested Reading

Suggested 'Starters' for SEM

- Kline, R. B. (2023). *Principles and practice of structural equation modeling* (5th ed.). The Guilford Press.
- Hoyle, R. K. (2023). *Handbook of Structural Equation Modeling* (2nd ed.). The Guilford Press.
- Bollen, K. A. (2026). *Elements of Structural Equation Models (SEMs)*. Cambridge University Press
- Hancock, G. R., & Mueller, R. O. (2013). *Structural Equation Modeling: A second course*. IAP.

Old but gold - Still the best book on SEM mathematical bases

- Bollen, K. A. (1989). *Structural equations with latent variables*. Wiley.

Longitudinal SEM

- Bollen, K. A., & Curran, P. J. (2006). *Latent curve models: A structural equation perspective*. Wiley.
- Little, T. D. (2024). *Longitudinal structural equation modeling* (2nd ed.). The Guilford Press.
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- Newsom, J. T. (2024). *Longitudinal structural equation modeling: A comprehensive introduction* (2nd ed.). Routledge.
- Grimm, K. J., Ram, N., & Estabrook, R. (2017). *Growth Modeling: Structural equation and multilevel modeling approaches*. The Guilford Press.
- Ferrer, E., Boker, S. M., & Grimm, K. J. (2019). *Longitudinal multivariate psychology*. Routledge.

Categorical Data

- Grimm, K. J. (2026). *Categorical Data Analysis with Structural Equation Models: Applications in Mplus and lavaan*. Guilford Press.



Suggested Reading

CFA

- Brown, T. A. (2015). *Confirmatory Factor Analysis for applied research* (2nd ed.). The Guilford Press.

Mixture Modeling (LCA, LPA, LTA)

- <https://mixture-modeling.netlify.app/>
- Nylund, K. L., Asparouhov, T., & Muthén, B. O. (2007). Deciding on the number of classes in latent class analysis and growth mixture modeling: A Monte Carlo simulation study. *Structural Equation Modeling: A Multidisciplinary Journal*, 14, 535-569. doi:10.1080/10705510701575396
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- van De Schoot, R., Sijbrandij, M., Winter, S. D., Depaoli, S., & Vermunt, J. K. (2017). The GRoLTS-Checklist: Guidelines for Reporting on Latent Trajectory Studies. *Structural Equation Modeling: A Multidisciplinary Journal*, 24, 451-467. doi: 10.1080/10705511.2016.1247646

Suggested Reading

Dynamic Structural Equation Modeling (DSEM)

- Asparouhov, T., Hamaker, E. L., & Muthén, B. (2018). Dynamic Structural Equation Models. *Structural Equation Modeling: A Multidisciplinary Journal*, 25(3), 359–388. <https://doi.org/10.1080/10705511.2017.1406803>
- Asparouhov, T., & Muthén, B. (2020). Comparison of models for the analysis of intensive longitudinal data. *Structural Equation Modeling: A Multidisciplinary Journal*, 27(2), 275–297. <https://doi.org/10.1080/10705511.2019.1626733>
- Hamaker, E. L., Asparouhov, T., Brose, A., Schmiedek, F., & Muthén, B. (2018). At the frontiers of modeling intensive longitudinal data: Dynamic Structural equation models for the affective measurements from the COGITO study. *Multivariate Behavioral Research*, 53(6), 820–841. <https://doi.org/10.1080/00273171.2018.1446819>
- Hamaker, E. L., Asparouhov, T., & Muthén, B. (2023). Dynamic structural equation modeling as a combination of time series modeling, multilevel modeling, and structural equation modeling. In R. H. Hoyle (Ed.), *Handbook of structural equation modeling* (2nd ed., pp. 576-596). The Guilford Press.
- McNeish, D., & Hamaker, E. L. (2020). A primer on two-level dynamic structural equation models for intensive longitudinal data in Mplus. *Psychological Methods*, 25(5), 610–635. <https://doi.org/10.1037/met0000250>
- Wang, J., & Wang, X. (2020). *Structural Equation Modeling: Applications Using Mplus* (2nd ed.). John Wiley & Sons. [Section 4.8, pp. 241-252]



Suggested Reading

Maximum Likelihood

- Enders, C. K. (2022). *Applied missing data analysis* (2nd ed.)(Chapter 2). The Guilford Press.
- Myung, I. J. (2003). Tutorial on maximum likelihood estimation. *Journal of Mathematical Psychology*, 47, 90-100. doi: 10.1016/S0022-2496(02)00028-7
- Stigler, S. M. (2007). The epic story of maximum likelihood. *Statistical Science*, 22(4), 598-620.
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[https://en.wikipedia.org/wiki/Maximum likelihood estimation](https://en.wikipedia.org/wiki/Maximum_likelihood_estimation)

Mplus

- Geiser, C. (2013). *Data analysis with Mplus*. The Guilford Press.
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- Byrne, B. M. (2013). *Structural equation modeling with Mplus: Basic concepts, applications, and programming*. Routledge.
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- Wickrama, K. A. S., Lee, T. K., O'Neal, C. W., & Lorenz, F. O. (2022). *Higher order growth curves and mixture modeling with Mplus: A practical guide* (2nd ed.). Routledge.
- Heck, R. H., & Thomas, S. L. (2015). *An introduction to multilevel modeling techniques: MLM and SEM approaches using Mplus* (3rd ed.). Routledge.
- Muthén, B., Muthén, L. K., Asparouhov, T. (2016). *Regression and mediation analysis using Mplus*. Muthén & Muthén.



Suggested Reading

lavaan (R package)

- <https://lavaan.ugent.be/tutorial/>
- <https://lavaan.ugent.be/tutorial/tutorial.pdf>
- <https://stats.oarc.ucla.edu/r/seminars/rsem/>
- https://users.ugent.be/~yrosseel/lavaan/gent2020/lavaan_twodays_gent2020.pdf
- Geiser, C. (2023). Structural equation modeling with the *Mplus* and lavaan programs. In R. Hoyle (Ed.), *Handbook of structural equation modeling* (2nd ed., pp. 241-258). Guilford Press.

SEM in Python with semopy package

- <https://pypi.org/project/semopy/> and <https://semopy.com/>
- Igolkina, A. A., & Meshcheryakov, G. (2020). semopy: A Python package for Structural Equation Modeling. *Structural Equation Modeling: A Multidisciplinary Journal*, 27(6), 952-963. <https://doi.org/10.1080/10705511.2019.1704289>

Bayesian SEM (BSEM)

Books and Chapters

- Depaoli, S. (2021). *Bayesian Structural Equation Modeling*. The Guilford Press.
- Kaplan, D. (2014). *Bayesian statistics for the social sciences*. The Guilford Press.
- Kaplan, D., & Depaoli, S. (2012). Bayesian Structural Equation Modeling. In R. H. Hoyle (Ed.), *Handbook of Structural Equation Modeling* (pp. 650-673). The Guilford Press.
- Lee, S. Y., & Song, X. Y. (2012). *Basic and advanced Bayesian structural equation modeling: With applications in the medical and behavioral sciences*. John Wiley & Sons.



Suggested Reading

Bayesian SEM (BSEM)

Articles:

- Lee, S. Y. (1981). A Bayesian approach to confirmatory factor analysis. *Psychometrika*, 46(2), 153-160. [First BSEM ever]
- Depaoli, S., & van de Schoot, R. (2017). Improving transparency and replication in Bayesian statistics: The WAMBS-Checklist. *Psychological Methods*, 22(2), 240–261.
- McNeish, D. (2016) On using bayesian methods to address small sample problems. *Structural Equation Modeling: A Multidisciplinary Journal*, 23(5), 750-773.
- Muthén, B., & Asparouhov, T. (2012). Bayesian structural equation modeling: A more flexible representation of substantive theory. *Psychological Methods*, 17(3), 313–335.
- Smid, S. C., McNeish, D., Miočević, M., & van de Schoot, R. (2020). Bayesian versus frequentist estimation for Structural Equation Models in small sample contexts: A systematic review. *Structural Equation Modeling: A Multidisciplinary Journal*, 27(1), 131-161.
- van de Schoot, R., Kaplan, D., Denissen, J., Asendorpf, J. B., Neyer, F. J., & Van Aken, M. A. (2014). A gentle introduction to Bayesian analysis: Applications to developmental research. *Child Development*, 85(3), 842-860.
- Zondervan-Zwijnenburg, M., Peeters, M., Depaoli, S., & van de Schoot, R. (2017). Where do priors come from? Applying guidelines to construct informative priors in small sample research. *Research in Human Development*, 14(4), 305-320.



Suggested Reading

Machine Learning and Latent Variables/Psychometrics (*Part 1*)

- Bleidorn, W., & Hopwood, C. J. (2019). Using machine learning to advance personality assessment and theory. *Personality and Social Psychology Review*, 23(2), 190-203. <https://doi.org/10.1177/1088868318772990>
- Brandmaier, A. M., & Jacobucci, R. C. (2023). Machine Learning approaches to structural Equation Modeling. In R. H. Hoyle (Ed.), *Handbook of structural equation modeling* (2nd ed., pp. 722-739). The Guilford Press.
- Brandmaier, A. M., von Oertzen, T., McArdle, J. J., & Lindenberger, U. (2013). Structural equation model trees. *Psychological Methods*, 18(1), 71–86. <https://doi.org/10.1037/a0030001>
- Grimm, K. J., Mazza, G. L., & Davoudzadeh, P. (2017). Model selection in finite mixture models: A *k*-fold cross-validation approach. *Structural Equation Modeling: A Multidisciplinary Journal*, 24(2), 246-256. <https://doi.org/10.1080/10705511.2016.1250638>
- Hong, M., Jacobucci, R., & Lubke, G. (2020). Deductive data mining. *Psychological Methods*, 25(6), 691–707. <https://doi.org/10.1037/met0000252>
- Jacobucci, R., Brandmaier, A. M., & Kievit, R. A. (2019). A practical guide to variable selection in Structural Equation Modeling by using regularized multiple-indicators, multiple-causes models. *Advances in Methods and Practices in Psychological Science*, 2(1), 55-76. <https://doi.org/10.1177/2515245919826527>
- Jacobucci, R., & Grimm, K. J. (2020). Machine learning and psychological research: The unexplored effect of measurement. *Perspectives on Psychological Science*, 15(3), 809-816. <https://doi.org/10.1177/1745691620902467>
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- Jacobucci, R., Grimm, K. J., & Zhang, Z. (2023). *Machine learning for social and behavioral research*. Guilford.
- Landers, R. N. (n.d.). *Data science for social scientists*. <https://datascience.tntlab.org/>



Suggested Reading

Machine Learning and Latent Variables/Psychometrics (*Part 2*)

- Liang, X., & Jacobucci, R. (2020). Regularized structural equation modeling to detect measurement bias: Evaluation of lasso, adaptive lasso, and elastic net. *Structural Equation Modeling: A Multidisciplinary Journal*, 27(5), 722-734. <https://doi.org/10.1080/10705511.2019.1693273>
- Liem, C. C. S., Langer, M., Demetriou, A., Hiemstra, A. M. F., Wicaksana, A. S., Born, M. P., & König, C. J. (2018). Psychology meets machine learning: Interdisciplinary perspectives on algorithmic job candidate screening. In H. J. Escalante, S. Escalera, I. Guyon, X. Baró, Y. Güçlütürk, U. Güçlü, & M. van Gerven (Eds.), *Explainable and interpretable models in computer vision and machine learning* (pp. 197-253). Springer. https://doi.org/10.1007/978-3-319-98131-4_9
- McNeish, D. M. (2015). Using lasso for predictor selection and to assuage overfitting: A method long overlooked in behavioral sciences. *Multivariate Behavioral Research*, 50(5), 471-484. <https://doi.org/10.1080/00273171.2015.1036965>
- Orrù, G., Monaro, M., Conversano, C., Gemignani, A., & Sartori, G. (2020). Machine learning in psychometrics and psychological research. *Frontiers in Psychology*, 10, Article 2970. <https://doi.org/10.3389/fpsyg.2019.02970>
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- Yarkoni, T., & Westfall, J. (2017). Choosing prediction over explanation in psychology: Lessons from machine learning. *Perspectives on Psychological Science*, 12(6), 1100-1122. <https://doi.org/10.1177/1745691617693393>



Suggested Reading

A whole journal devoted to the advancement of SEM (since 1994)

Taylor & Francis Online

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And those “latent variables,” are they in the room with us right now?



71

1.272

9.060





**Thanks for
your attention**

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